

Article

A Two-Stage Coordinated Dispatch Framework for Integrated Energy Systems with Growing Wind Power Penetration Considering Price-Based Demand Response

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Abstract

With the strategic advancement of energy structure transformation and the implementation of carbon peaking and carbon neutrality goals, the Integrated Energy System (IES) has become a core research direction owing to its superior performance in multi-energy complementation, operational efficiency, and low-carbon emission characteristics. Nevertheless, existing studies reveal that the optimal operation of IES still faces significant challenges, including the high complexity of multi-energy coupling, supply–demand imbalance caused by renewable energy penetration, and insufficient exploitation of demand-side flexibility. As a core measure of demand-side management, demand response (DR) provides an effective approach to motivate users to adjust power load via price incentives or direct load control. DR can effectively smooth load profiles, improve resource utilization, and boost the consumption level of renewable energy. To meet the operational demands of modern IES, this paper establishes a security-constrained economic dispatch model embedded with multi-level demand response mechanisms. The proposed framework is divided into four key modules: First, a price-based demand response strategy is developed to dynamically guide users in regulating multi-energy consumption behaviors. Second, electric vehicles (EVs) are considered flexible demand-side resources with unique response characteristics. An aggregated EV charging–discharging model is established to suppress power fluctuations and support high proportions of renewable energy integration. Third, to precisely calculate the overall operating cost of IES, a combined economic evaluation index integrating time-of-use tariff and Levelized Cost of Electricity is adopted. It maintains a balance between amortized long-term generation investment and short-term operational expenditure, and coordinates the economic benefits and operational reliability of the whole system. Finally, numerical simulations are performed on a coupled test system comprising an IEEE 33-bus distribution network and a 20-node natural gas network. Simulation results verify that the proposed co-optimization model can effectively reduce total system operating costs and greatly improve the local assumption of fluctuating renewable energy.



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1. Introduction

Driven by the goals of carbon peaking and carbon neutrality, the global energy structure is undergoing a profound transition from a centralized fossil-fuel-dominated paradigm

to a decarbonized, distributed, and multi-vector configuration [1]. Against this backdrop, the Integrated Energy System (IES) has gained traction as an infrastructure that ties electricity, natural gas, heat, and cooling into one framework; because these carriers are coupled and energy can be converted across them, resources are used in cascade rather than carrier by carrier [2]. Compared with conventional decoupled energy networks, the IES offers significant advantages in accommodating renewable energy, achieving comprehensive energy efficiency, and reducing carbon emissions [3]. However, the high proportion of integration of fluctuating renewables, the strong nonlinear coupling among diverse energy carriers, and the insufficient exploitation of demand-side flexibility have posed significant challenges to the secure and economic operation of the IES [4]. Therefore, developing a coordinated co-optimization scheduling framework that fully unlocks supply-side and demand-side flexibility has become a research focus in recent years [5].

As a core measure of demand-side management, demand response (DR) provides an effective approach to motivate end users to actively regulate their energy consumption through price incentives or direct load control [6]. Among various DR mechanisms, price-based demand response (PDR) guided by time-of-use (TOU) tariffs has been widely adopted owing to its market compatibility, minimal administrative intervention, and broad social acceptability [7]. Reference [8] established a TOU pricing model based on a cross-price elasticity matrix to characterize consumers' multi-period, interdependent response behaviors. In reference [9], a multi-energy PDR framework was developed for an IES operator, in which shiftable and interruptible loads were jointly scheduled to flatten the net-load curve. Reference [10] proposed a dynamic TOU pricing mechanism that accounts for both wholesale procurement cost volatility and peak-to-valley regulation, thereby improving peak shaving performance. A Stackelberg game between the operator and multi-energy consumers was solved in [11] to recover the optimal retail price vector, while [12] folded renewable forecast uncertainty into a real-time pricing scheme so that flexible loads would respond more sharply to scarcity. Taken together, these works confirm that PDR can reshape the load profile. What they share is a narrow treatment of the TOU price vector: it is bounded but otherwise left free, and the constraints that keep the market itself stable—caps on the peak-to-valley ratio, an enforced ordering of peak, flat, and valley tariffs—are rarely written into the formulation, even though it is precisely these limits that protect social welfare and head off clearing failures.

Alongside these stationary resources, the spread of electric vehicles (EVs) has turned the vehicle-to-grid (V2G) concept into a fast-acting source of flexibility [13]. With bidirectional chargers, an aggregated fleet behaves like distributed mobile storage: it soaks up surplus renewable output overnight and returns energy to the grid when demand peaks [14]. A series of modeling efforts builds on this idea. The aggregated dispatch model in [15] represents the random arrival and departure of vehicles with normal distributions; ref. [16] coordinates charging to damp wind fluctuations and curb curtailment; ref. [17] studies V2G supplying ancillary services such as frequency regulation and spinning reserve inside a distribution network; and [18] embeds V2G in microgrid energy management to price the gains from two-way exchange. Comparing uncoordinated against coordinated operation, ref. [19] then quantifies how V2G smooths load and steadies voltage. What these studies have in common is that they stay almost entirely within the electrical subsystem, and seldom ask how V2G flexibility travels into a coupled carrier—above all how it relieves the gas network indirectly by trimming the peaking duty of gas turbines.

For the IES involving electricity and natural gas, cross-carrier coupling is primarily achieved through Combined Heat and Power (CHP) units, gas-fired turbines, and power-to-gas (P2G) facilities, which together form a bidirectional energy loop [20]. Reference [21] formulated a joint optimal power–gas flow model and demonstrated that multi-energy co-

ordination can effectively mitigate localized congestion in both networks. In reference [22], a P2G-based mechanism was developed to convert surplus wind generation into synthetic natural gas, achieving deep renewable accommodation. A security-constrained co-dispatch framework in [23] writes the Weymouth pipeline-flow equation directly into the problem, and [24] goes a step further, carrying the gas network's multi-period nodal pressures and line-pack behavior into a day-ahead schedule; ref. [25] then layers on renewable uncertainty and gas-source production limits. So, the physical electricity–gas coupling is modeled in steadily finer detail, yet the demand side stays underexplored—in particular, how PDR and V2G act together to relax the stress that accumulates between the two networks.

On the methodological side, IES scheduling is most often cast as a mixed-integer linear program (MILP), which solves reliably and runs on off-the-shelf commercial solvers [26]. Nonlinear terms—the Weymouth gas-flow relation and the quadratic generator cost curves—are handled through piecewise linearization, keeping the model within reach of those solvers [27]. From the economic evaluation perspective, conventional dispatch models often consider only short-term operational expenditures while neglecting the lifecycle degradation of capital-intensive assets such as electrochemical energy storage systems (EESS). To address this issue, the Levelized Cost of Electricity (LCOE) framework has been introduced to amortize lifecycle capital and maintenance expenditures over the asset operating horizon [28]. Reference [29] integrated LCOE-based depreciation costs for EESS into a day-ahead dispatch model to prevent overly optimistic charging–discharging schedules. Nevertheless, few studies have combined the LCOE evaluation with both TOU-based PDR and V2G coordination under a unified electricity–gas IES scheduling framework.

In response to the above analyses, this paper establishes a security-constrained economic co-optimization scheduling model for a coupled electricity–gas IES that embeds multi-level demand-side flexibility mechanisms. A TOU-based PDR strategy that considers cross-price elasticities and market stability constraints is developed to dynamically reshape multi-energy consumption profiles. An aggregated V2G model with stochastic mobility patterns is developed to exploit the bidirectional flexibility of EV fleets. A combined economic evaluation index integrating TOU tariff and LCOE is adopted to balance short-term operational expenditure and long-term asset depreciation. The main contributions of this paper are summarized as follows.

- (1) Under a coupled power–gas IES architecture, the synergistic interaction between PDR and V2G is comprehensively considered, which fully unlocks demand-side flexibility and coordinates the supply–demand balance across electricity and natural gas vectors.
- (2) The PDR framework rests on a TOU tariff and a cross-price elasticity description of user behavior, but unlike most formulations it is fenced in by explicit market-stability constraints: the peak, flat, and valley tariffs must keep their ordering, and the peak-to-valley ratio is held within fixed bounds. User adaptation is therefore captured mathematically without letting the pricing scheme distort market clearing.
- (3) A single cost index that pairs the TOU tariff with an LCOE term drives the joint dispatch of conventional units, P2G plants, and the EESS. Charging the storage for its lifecycle wear discourages the aggressive cycling that idealized models tend to produce, so the schedule settles on a workable balance between running cost and asset longevity.

To make the specific novelty of this work explicit relative to the studies reviewed above, the three contributions are positioned against representative prior art as follows. Unlike studies that treat price-based demand response, vehicle-to-grid operation, power-to-gas conversion, and storage-degradation costing in isolation or in pairwise combinations, the present framework embeds all of them within a single security-constrained electricity–gas co-optimization that simultaneously enforces distribution-network power

flow, gas-network Weymouth dynamics, and retail market-stability limits. The first novelty lies in coordinating demand-side price elasticity with bidirectional V2G inside the coupled power–gas network, so that demand reshaping and mobile storage are dispatched jointly rather than sequentially. The second novelty is the explicit incorporation of market-stability constraints—bounding the peak-to-valley ratio and the step change in the time-of-use tariff—directly inside the pricing model, which most prior TOU studies omit. The third novelty is the use of an LCOE-based degradation term that places storage and EV cycling on the same economic footing as fuel and procurement costs, discouraging the unrealistically aggressive cycling produced by degradation-free models. Table 1 contrasts these features against representative recent works.

Table 1. Comparison between the proposed framework and representative existing studies.

Feature	Typical DR/IES Studies	V2G/EV Studies	P2G–IES Studies	This Work
Price-based DR (TOU + elasticity)	partial	no	no	yes
Bidirectional V2G scheduling	no	yes	no	yes
P2G electricity–gas coupling	no	no	yes	yes
Storage/EV degradation via LCOE	no	partial	no	yes
Market-stability limits on TOU tariff	no	no	no	yes
Security-constrained power–gas co-optimization	partial	no	partial	yes

The rest of the paper proceeds as follows. Section 2 sets out how the IES operates under demand response, covering the operator architecture, the TOU pricing mechanism, and the PDR framework. The optimization model and its layered constraints are built up in Section 3. Section 4 reports a case study on a coupled IEEE 33-bus and 20-node gas network, and Section 5 draws the conclusions.

2. Operational Mechanisms for IES-DR

2.1. Operator Architecture and Energy Flow Model for IES

2.1.1. Structure of the IES Operator

The Integrated Energy System (IES), serving as a crucial physical paradigm of the modern Energy Internet, is an advanced energy infrastructure that organically synthesizes the production, transmission, storage, and consumption of diverse energy vectors—including electricity, thermal energy, natural gas, and cooling power—via highly interconnected physical distribution and transmission multi-networks, thereby achieving cross-carrier multi-energy synergy, cascade utilization, and low-carbon transformations. Inside this system, the IES Operator (IESO) is the entity that actually runs things: it manages the internal co-dispatch, decides how the multi-carrier coupling components are configured, and handles clearing and settlement in the retail energy markets. Its makeup is correspondingly broad, spanning source-side input interfaces, the cross-carrier conversion grids, hybrid storage banks, terminal load-control systems, and a Cyber-Physical Systems (CPS) management center. This is a departure from the older picture of an isolated power or gas dispatcher; the operator now aggregates several energy streams at once, converts among them, coordinates the network as a whole, and matches supply to end-use demand as conditions change. By dissolving the physical and managerial divisions that come with running each carrier on its own, the IESO becomes a single digital control entity with unified dispatch authority—which in turn hardens the system against the shocks of high renewable penetration and opens room to develop the multi-energy market.

2.1.2. Analysis of Internal Energy Flow and Coupling Characteristics

Complex nonlinear coupling and multi-form transformations among electricity, heat, gas, and cooling vectors across spatiotemporal scales interweave within the IES's internal architecture as shown as Figure 1. On the energy input side, the IES exhibits strong inclusivity, accommodating diverse multi-energy streams from upstream wholesale markets and natural energy networks. These include highly stochastic, volatile clean power from wind turbines and photovoltaic arrays; thermal energy harvested from gas boilers; deep geothermal resources; and industrial waste heat recovery systems, as well as natural gas sourced from primary high-pressure pipelines. During the internal conversion stage, various multi-energy converters serve as physical nodes for energy transmission. Take electric heat pumps (HPs): their high coefficient of performance (COP) lets high-grade electricity be turned into heat or cooling at low loss. Power-to-gas (P2G) units run in the opposite direction, drawing on surplus or cheaply curtailed wind and solar to electrolyze water and convert the resulting hydrogen, which opens a reverse path from electricity back to gas and lets energy move across the two grids. Combined Heat and Power (CHP) units consume natural gas to generate high-grade electricity and usable thermal energy simultaneously, serving as core devices for multi-energy coupling. Absorption chillers further recover high-temperature waste heat from CHPs or boilers to produce cooling energy, realizing cascade utilization of thermal energy for cooling supply. On the energy storage and regulation side, the system organically integrates a hybrid energy storage system comprising lithium-ion batteries, vanadium redox flow batteries, hot water tanks, ice storage tanks, and underground gas storage caverns, thereby extending the temporal flexibility of energy dispatch. On the terminal demand side, load profiles cover highly interruptible heavy industrial loads, shiftable thermal and cooling loads with strong thermal inertia, and spatially movable Vehicle-to-Grid (V2G) bidirectional charging piles. The inherent temporal inertia of thermal and gas energy, coupled with multi-dimensional spatial energy conversions, defines the physical boundaries and mathematical constraints for internal energy flow optimization [30].

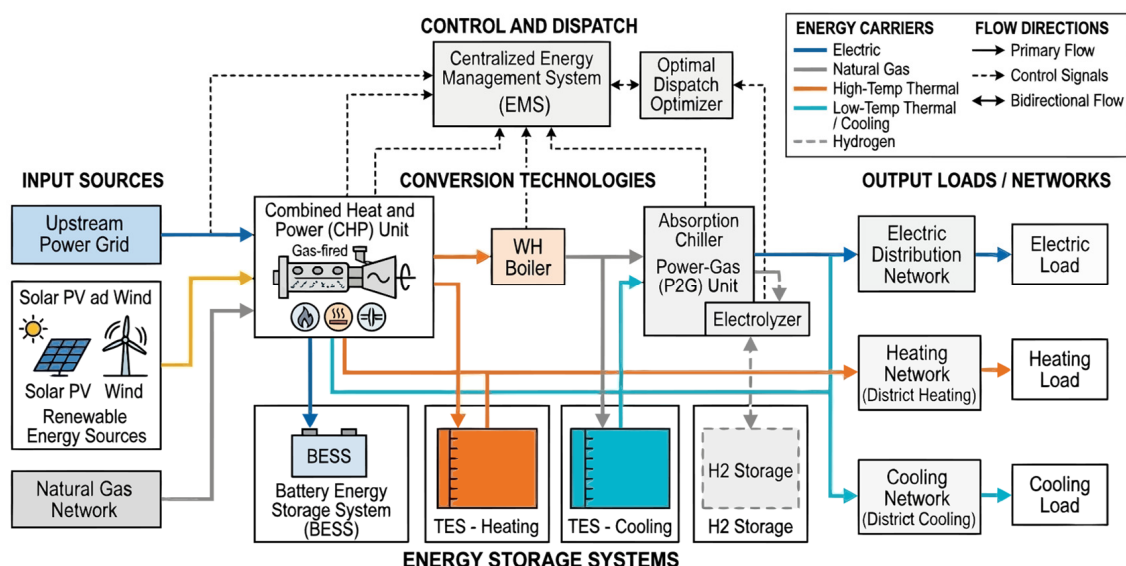


Figure 1. The structure of the IES.

2.2. TOU Pricing Mechanism Considering Energy Procurement Cost

2.2.1. Upstream Energy Procurement Cost Mechanism

The IES operator acts simultaneously as a wholesale-side buyer and a retail-side seller within the energy value chain. On the wholesale trade front, the operator faces

diverse procurement cost variations driven by external bulk power grid spot markets, long-term high-pressure natural gas pipeline contracts, and dynamic multi-energy commodity markets. Procurement on the power side is volatile, tracking day-ahead spot clearing prices, real-time balancing charges, and capacity-based network tariffs at the same time. When demand peaks or renewable output runs low, the day-ahead price climbs steeply; on windy nights, with wind plentiful, it collapses. Upstream natural gas procurement costs are governed by a hybrid weighted combination of long-term pipeline capacity contract prices and dynamic spot gas prices, exhibiting nonlinear stepped fluctuations subject to gas pipeline pressure constraints. To mathematically solve the system dispatch problem over a scheduling horizon T , the operator's comprehensive wholesale multi-energy procurement cost function is formulated as follows.

$$C_{proc} = \sum_{t=1}^T \left[\lambda_{e,spot}^t P_{grid}^t + \lambda_{g,spot}^t V_{gas}^t + \lambda_{h,wholesale}^t Q_{heat}^t + C_{punish}(P_{curtail}^t) + \sum_{m \in MC} C_{m,O\&M}(P_m^t) \right] \quad (1)$$

where C_{proc} is the total multi-energy procurement and systemic operational cost of the operator across the entire scheduling horizon T ; $\lambda_{e,spot}^t P_{grid}^t$ represents the cost of electricity purchased from the external grid, where $\lambda_{e,spot}^t$ is the dynamic clearing tariff of the upstream wholesale power spot market at time slot t and P_{grid}^t is the active power procured by the operator from the external bulk grid; $\lambda_{g,spot}^t V_{gas}^t$ represents the cost of purchased natural gas, where $\lambda_{g,spot}^t$ is the dynamic gas price of the upstream natural gas network at time slot t and V_{gas}^t is the volumetric flow rate of natural gas consumed by the system; $\lambda_{h,wholesale}^t Q_{heat}^t$ represents the cost of purchased thermal energy, where $\lambda_{h,wholesale}^t$ is the wholesale heat price of the upstream district heating network at time slot t and Q_{heat}^t is the actual thermal power procured; $C_{punish}(P_{curtail}^t)$ denotes the penalty cost for wind and solar curtailment, which is formulated as a high-penalty function to enforce the prioritized accommodation of local renewable energy and mitigate economic losses incurred by unabsorbed wind/solar output $P_{curtail}^t$; $\sum_{m \in MC} C_{m,O\&M}(P_m^t)$ is the total operation, maintenance, and depreciation cost of internal coupling devices, where M represents the set of all multi-energy conversion components, and $C_{m,O\&M}$ is the comprehensive O&M and physical wear-and-tear depreciation cost function of the m -th device operating at an output level of P_m^t .

2.2.2. Downstream TOU-Based Load Regulation Framework for Operators

To efficiently transfer wholesale-side spot price volatility, operational cost pressures, and physical renewable supply–demand imbalances down to terminal consumption layers—while unleashing the latent cross-carrier flexibility of multi-energy complementary optimization—the operator must construct an adaptive retail-side TOU pricing and multi-energy load regulation framework. This framework centers on customized retail TOU electricity tariffs, accompanied by temporally coordinated retail TOU gas and thermal price matrices. By deep mining historical multi-energy load big data and upstream spot-clearing patterns, the operator segments the 24 h horizon into peak, flat, and valley blocks. Daytime hours, when commercial load and procurement cost are both high, fall into the peak band; the quiet overnight hours, when wind is abundant, become the valley. Widening the gap between the two prices is what gives the tariff its bite. A high peak rate, switched on when bulk power is tight or procurement turns expensive, pushes users to trim flexible load or postpone it, while a low valley rate, released during heavy curtailment or the overnight trough, pulls demand the other way. Underneath sits the usual digital layer—smart grids, an automated Advanced Distribution Management System, an Energy Management System (EMS), and dense Advanced Metering Infrastructure (AMI)—which tracks the power, heat, and gas use of each user in a closed loop and keeps the physical, information, and price flows tied together for the sake of flexibility and grid stability.

2.3. TOU-Based PDR Framework

Putting price-based demand response (PDR) into practice inside the IES means coordinating flexible resources through prices rather than commands: the operator sets multi-carrier retail price vectors and lets the market do the balancing. The idea rests on one mathematical object, the cross-price elasticity matrix, which encodes how end-users respond economically to those prices. Demand response then closes a loop between grid and consumer, and the price elasticity of demand is what measures the strength of that two-way link.

In concrete terms, the price elasticity of demand measures how far load moves for a given change in commodity price over a period, and is written as

$$\varepsilon = \frac{\Delta P / P_0}{\Delta \lambda / \lambda_0} \quad (2)$$

where ε denotes the price elasticity coefficient; ΔP and $\Delta \lambda$ represent the variations in load demand and retail price before and after the execution of demand response, respectively; P_0 and λ_0 are the baseline load demand and the initial energy price prior to demand response, respectively.

In practical power grid operations, load variation within a specific time interval not only correlates with concurrent prices but is also inherently influenced by energy prices across other operational intervals throughout the day. To accurately capture these multi-period interdependencies of a single energy vector across the scheduling horizon, the cross-price elasticity coefficient is introduced. The cross-price elasticity of demand reflects the sensitivity of the demand quantity in a given period to the price variations in another correlated period, expressed as

$$\varepsilon_{ij} = \frac{\Delta P_i / P_{i,0}}{\Delta \lambda_j / \lambda_{j,0}} \quad (3)$$

where ε_{ij} denotes the cross-price elasticity coefficient between time slot i and time slot j ; ΔP_i is the variation in load demand during time slot i ; $\Delta \lambda_j$ signifies the price change during time slot j ; $P_{i,0}$ represents the initial baseline load in time slot i ; and $\lambda_{j,0}$ is the initial electricity price in time slot j .

Over a scheduling horizon of n periods, the collective response of end-users to the price vector collapses into a single object, the demand price elasticity matrix $\mathbf{E}$:

$$\mathbf{E} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \dots & \varepsilon_{1n} \\ \varepsilon_{21} & \varepsilon_{22} & \dots & \varepsilon_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{n1} & \varepsilon_{n2} & \dots & \varepsilon_{nn} \end{bmatrix} \quad (4)$$

Once the operator derives multi-carrier TOU or real-time pricing (RTP) signals from its wholesale cost analytics and pushes them over a two-way communication network to the retail nodes, each user responds on its own terms—minimizing its bill—and the sum of those self-interested adjustments reshapes the multi-energy demand profile, so peak shaving happens without anyone being told to do it.

To represent this adjustment, loads are split into three groups according to how freely they can be moved or shed: fixed, shiftable, and interruptible.

Fixed loads are the baseline demand that cannot be moved in time or trimmed in magnitude. Supply to them has to be continuous, since interrupting it would disrupt production or daily life, and because users will not—or cannot—change how they draw

this power, the quantity holds essentially flat even when prices spike. In the PDR model they enter as demand that must be met first, defined as

$$P_{fixed}^t = C_{fixed} \quad (5)$$

where P_{fixed}^t denotes the fixed load at time slot t , and C_{fixed} is a constant representing the rigid demand threshold.

Shiftable loads keep the same total energy over the cycle but can be slid along the time axis. Responding to retail prices, users bring these operations forward or push them back, and because such customers are price-sensitive they tend to migrate consumption into the cheap valley hours, sometimes backed by on-site storage. The model for shiftable load is

$$P_{shift}^t = P_{shift}^{t,0} + \sum_{j=1}^n \varepsilon_{tj,shift} \cdot P_{shift}^{t,0} \cdot \frac{\lambda^j - \lambda^{j,0}}{\lambda^{j,0}} \quad (6)$$

$$\sum_{t=1}^n P_{shift}^t = \sum_{t=1}^n P_{shift}^{t,0} \quad (7)$$

$$0 \leq P_{shift}^t \leq P_{shift}^{max} \quad (8)$$

where P_{shift}^t and $P_{shift}^{t,0}$ represent the post-response and pre-response shiftable loads at time slot t , respectively; $\varepsilon_{tj,shift}$ is the element of the demand price elasticity matrix reflecting the temporal shifting characteristics of the load; λ^j and $\lambda^{j,0}$ denote the post-response and pre-response electricity prices at time slot j , respectively; and P_{shift}^{max} represents the upper capacity limit of shiftable loads that can be accommodated within the given time interval.

Interruptible loads can be cut back or switched off for a while. Under a prior contract or a compensation arrangement, the user lets the operator suspend supply when defined conditions arise; such loads are usually energy-intensive and can ride through short outages or reduced-power spells, which makes them well suited to emergency frequency support or operating reserve. The interruptible-load model reads

$$P_{inter}^t = P_{inter}^{t,0} + \sum_{j=1}^n \varepsilon_{tj,inter} \cdot P_{inter}^{t,0} \cdot \frac{\lambda^j - \lambda^{j,0}}{\lambda^{j,0}} \quad (9)$$

$$0 \leq \Delta P_{inter}^t \leq P_{inter}^{max} \quad (10)$$

where P_{inter}^t and $P_{inter}^{t,0}$ represent the post-response and pre-response interruptible loads at time slot t , respectively; $\varepsilon_{tj,inter}$ is the corresponding elasticity matrix element mapping the curtailment behavior; and P_{inter}^{max} denotes the maximum allowable interruption capacity.

To guarantee that flexible resources are not counted twice, the baseline demand at each node is partitioned into three mutually exclusive and collectively exhaustive subsets, namely fixed, shiftable, and interruptible, so that any given unit of load belongs to exactly one demand response category. A single consumer block is therefore prevented by construction from participating simultaneously in both the shiftable and the interruptible program, and the post-response load is obtained by summing over these disjoint subsets. This partitioning removes any possibility of double-counting the same flexibility across two mechanisms.

Consequently, by aggregating the aforementioned distinct load components, the total post-demand response load profile P_{total}^t interacting with the power grid can be dynamically derived as

$$P_{total}^t = P_{fixed}^t + P_{shift}^t + P_{inter}^t \quad (11)$$

where P_{total}^t represents the modified aggregate load of the grid at time slot t following the implementation of PDR.

On a practical spatiotemporal operational plane, this multi-energy response manifests through three highly complementary dimensions:

1. Temporal shifting and adaptive curtailment of flexible multi-carrier loads: Upon observing premium retail peak power tariffs during industrial peak hours, large-scale industrial microgrids automatically reduce the direct current of their smelting cells; commercial facilities simultaneously utilize the structural thermal inertia of modern smart buildings to actively scale down HVAC compressor power during peak intervals, smoothly deferring these large thermal-driven power loads to nighttime valley periods with ultra-low energy prices.
2. Cross-carrier substitution between energy vectors: Where heat can be met either electrically or by gas, the choice follows price. If daytime power prices jump while gas tariffs hold steady, a dual-fuel consumer backs off its electric heat pumps and heats with a gas boiler or CHP instead; when wind floods the system overnight and power turns cheap, the same consumer drops gas and heats electrically. The swap is horizontal, carrier for carrier, and driven by the retail tariff alone.

The cross-price elasticity matrix is not estimated afresh in this study; its self- and cross-elasticity entries are adopted from the established demand response literature for electricity end-uses and are treated as exogenous parameters representative of a price-responsive commercial, residential mix, which an operator would in practice calibrate from historical smart-meter and tariff records by regressing observed load on price. Realism and user acceptability are preserved not through the elasticity values alone but through the bounds imposed on the response. User comfort and the limited willingness to participate are captured by capping the shiftable and interruptible quantities at fixed fractions of the nodal baseline load, so that only a contractually realistic share of demand can be moved or curtailed in any period. The rebound effect is bounded implicitly by the daily energy-conservation constraint on shiftable load, which forces demand deferred out of peak hours to be recovered within the same day without creating a new artificial peak. Interruptible-load compensation is reflected through the curtailment cost the operator pays in the objective, so that interruption is invoked only when it is economically justified. Multi-energy substitution is likewise bounded by the conversion-capacity limits of the coupling devices, preventing unrealistic instantaneous switching between energy vectors. Together these bounds keep the modeled demand response within a physically and behaviorally plausible envelope.

3. Arbitrage in time through on-site storage and V2G: Behind-the-meter storage buys low and sells high, charging through the cheap valley and discharging into the expensive peak. EV fleets parked across the network do the same through bidirectional chargers, taking in surplus wind and solar overnight and returning energy during the worst of the peak, and they pocket the peak–valley spread for it.

Because PDR works through price signals and the independent choices users make in response, it needs no administrative compulsion. That keeps it flexible, low on friction, and easy to accept across residential and commercial districts. As energy networks couple more tightly and shift toward a low-carbon footing, a cross-elasticity TOU scheme of this kind is likely to matter more rather than less: it helps absorb large, variable amounts of renewable power, leaves the system better able to withstand cascading failures, cuts avoidable fossil use, and moves the carbon-neutrality target closer.

3. Mathematical Modeling and Constraints

To prevent any ambiguity arising from the term “two-stage,” the proposed framework is organized into two sequentially coupled decision stages that share the same 24 h day-ahead horizon at hourly resolution but differ in their decision variables and information sets, as summarized in Figure 2. In the first stage, the retail-pricing and demand-shaping stage, the IES operator determines the time-of-use retail price vector for the peak, flat, and valley blocks subject to the market-stability constraints of Section 3.2.1; acting through the cross-price elasticity matrix, this price vector yields the price-responsive shiftable and interruptible load. The settled price vector and the resulting post-PDR nodal load are the first-stage decision variables and are passed downward as fixed boundary conditions. In the second stage, the security-constrained dispatch stage, taking the reshaped load as the boundary condition, the operator co-optimizes the dispatch of conventional and gas-fired units, CHP and P2G plants, the EESS, and the bidirectional V2G fleet, together with the coupled electricity- and gas-network constraints, so as to minimize the total operating cost. The two stages are coupled through a one-directional price-to-load information flow: the first stage decides how much demand to move and at what price, and the second stage decides how to serve the resulting demand securely and economically. Because the user response is endogenous to the pricing decision and both stages share the same day-ahead time scale, they are assembled and solved as a single mixed-integer linear programming (MILP) instance rather than by an iterative master–subproblem decomposition; this preserves global optimality while retaining the conceptual two-stage decision logic. The second stage therefore plays the role of the recourse-like dispatch layer that realizes the price decision rather than a separate real-time correction.

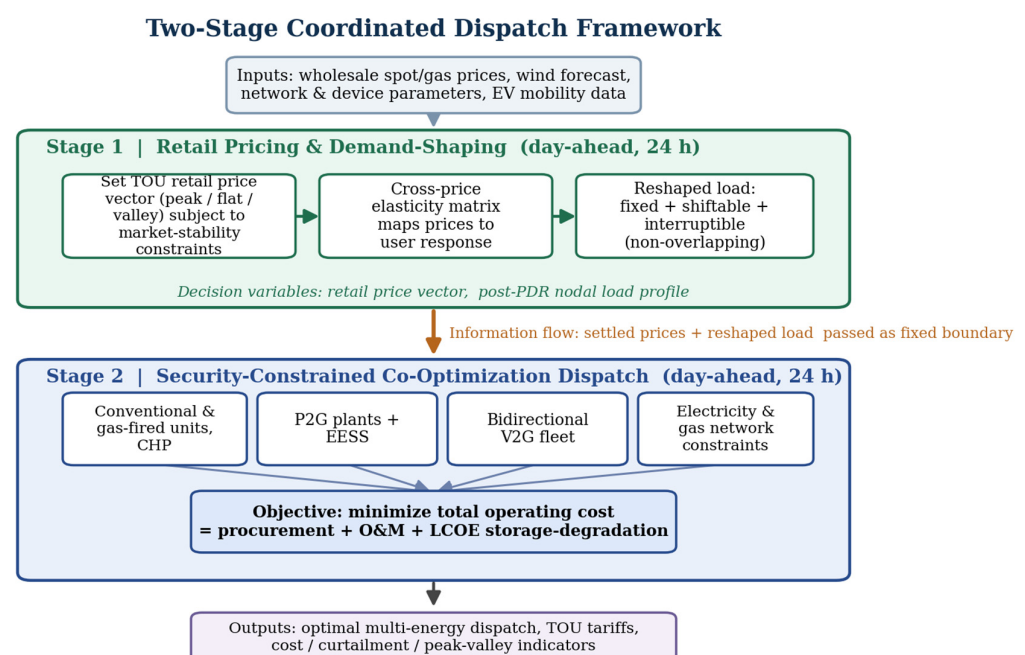


Figure 2. Information flow of the proposed two-stage coordinated dispatch framework.

3.1. Objective Function

To balance the systemic economic benefits with the complementary advantages of multi-carrier energy coupling, this study establishes an economic-optimal scheduling model with the core objective of minimizing the system’s total operational cost within a day-ahead scheduling horizon. This objective function not only coordinates the synergetic procurement of multi-carrier energy flows on the wholesale side but also mathematically captures the nonlinear energy-consumption characteristics of multi-energy conversion

devices and the lifecycle degradation of energy storage assets. The objective function to minimize the total systemic cost is formulated as follows:

$$\begin{aligned} \min C &= \min(C_1 + C_2 + C_3) \\ \left\{ \begin{aligned} C_1 &= \sum_{t \in T} \left(\lambda_{e,\text{spot}}^t P_{\text{grid}}^t + \lambda_{g,\text{spot}}^t V_{\text{gas}}^t \right) \\ C_2 &= \sum_{t \in T} \left[\sum_{h \in N_h} \left(a_h (P_{h,t})^2 + b_h P_{h,t} + c_h \right) + \sum_{u \in N_u} \left(a_u (P_{u,t})^2 + b_u P_{u,t} + c_u \right) + \sum_{p \in N_p} C_p P_{p,t} \right] \\ C_3 &= C_{\text{LCOE}} \sum_{t \in T} P_{\text{EESS},t}^{\text{out}} \Delta t \end{aligned} \right. \quad (12) \end{aligned}$$

where C is the total day-ahead operational cost of IES over the scheduling horizon and C_1 , C_2 , and C_3 represent the upstream multi-energy procurement cost, the equipment operations and maintenance (O&M) cost, and the electrochemical energy storage system (EESS) lifecycle depreciation cost, respectively.

The upstream multi-energy procurement cost C_1 quantifies the total capital expenditure incurred by the IES operator when purchasing primary energy from external multi-carrier energy networks within the day-ahead wholesale market. By embedding time-varying price vectors, this term guides the system to perceive external market price fluctuations at the source side. Its mathematical expression is given as above, where T represents the complete time horizon for day-ahead economic scheduling; in this study, $T = 24$ h with a discrete operational time step of $\Delta t = 1$ h; $\lambda_{e,\text{spot}}^t$ and $\lambda_{g,\text{spot}}^t$ denote the dynamic clearing tariff of the upstream power spot market (RMB/kWh) and the real-time procurement price of the natural gas market (RMB/m³) at time slot t , respectively; P_{grid}^t is the active power exchanged with the external main grid through the tie-line at time slot t (kW), where a positive value signifies power procurement; and V_{gas}^t represents the volumetric flow rate of natural gas purchased from the external gas source at time slot t (m³/h).

The equipment operations and maintenance cost C_2 is designed to rigorously quantify the fuel consumption, routine maintenance, and physical wear-and-tear of core energy-conversion units within the IES during the execution of horizontal multi-energy complementation and temporal energy shifting. Specifically, the energy consumption characteristics of gas-fired power plants and CHP units are approximated using quadratic functions. This nonlinear formulation accurately models the real-world physical behavior in which thermal prime movers deviate from their nominal efficiency at part-load conditions. Conversely, P2G facilities are characterized using a linear operational cost coefficient. The comprehensive cost function is expressed as above, where N_h and N_u represent the sets of internal CHP units and conventional gas-fired power plants, respectively; N_p denotes the set of P2G conversion devices; $P_{h,t}$ and $P_{u,t}$ indicate the actual active power output of the h -th CHP unit and the u -th gas-fired plant at time slot t (kW), respectively; $P_{p,t}$ represents the active electrical power input consumed by the p -th P2G device at time slot t (kW); a_h, b_h, c_h and a_u, b_u, c_u are the quadratic, linear, and constant cost coefficients governing the fuel consumption and O&M expenses of the CHP and gas plants, respectively; and C_p signifies the unit O&M cost coefficient of the P2G facilities (RMB/kWh).

The electrochemical energy storage system (EESS) lifecycle cost explicitly maps the implicit costs of physical degradation, cycle-life degradation, and internal chemical efficiency losses using Levelized Cost of Energy (LCOE) principles, thereby preventing aggressive or overly optimistic scheduling profiles resulting from idealized modeling. This cost component is quasi-linearized based on the discharge throughput over the dispatch horizon, where C_{LCOE} represents the levelized depreciation and maintenance cost coefficient of the EESS (RMB/kWh), which comprehensively factors in the initial capital investment, battery replacement costs, maximum cycle life, and salvage value; and $P_{\text{EESS},t}^{\text{out}}$ denotes the actual active discharging power of the EESS at time slot t (kW). The integration of this

term effectively penalizes the “shallow cycle creeping” caused by frequent charging and discharging transitions, mitigating accelerated aging of battery assets and ensuring an optimal trade-off between day-ahead operational economy and asset longevity.

3.2. Constraints

3.2.1. TOU Pricing Vector and Market Stability Constraints

When designing or implementing retail TOU electricity pricing strategies, the grid operator must systematically balance the regulatory efficacy of grid peak shaving with social welfare and market financial sustainability. The peak tariff has to be high enough to hold back demand when it matters most, but push it too far and consumer goodwill, household budgets, and industrial output all suffer. The valley tariff faces the mirror problem: low enough to pull load into the surplus hours, yet not so low that upstream generators are starved of revenue. To keep the pricing vector from drifting into gouging, welfare loss, or outright clearing failure, it is hemmed in by a set of tiered bounds.

First, the hierarchical relationship and absolute operational boundaries governing the peak, flat, and valley electricity tariffs are mathematically constrained as follows:

$$\lambda_{\text{peak}}^{\min} \leq \lambda_e^t \leq \lambda_{\text{peak}}^{\max}, \forall t \in \Omega_{\text{peak}} \quad (13)$$

$$\lambda_{\text{valley}}^{\min} \leq \lambda_e^t \leq \lambda_{\text{valley}}^{\max}, \forall t \in \Omega_{\text{valley}} \quad (14)$$

$$\max_{t \in \Omega_{\text{valley}}} (\lambda_e^t) < \lambda_e^\tau < \min_{t \in \Omega_{\text{peak}}} (\lambda_e^t), \forall \tau \in \Omega_{\text{flat}} \quad (15)$$

where Ω_{peak} , Ω_{flat} , and Ω_{valley} denote the mutually exclusive sets of peak, flat, and valley time slots within the scheduling horizon T , respectively; λ_e^t and λ_e^τ represent the optimized retail electricity tariffs at the corresponding time intervals; $\lambda_{\text{peak}}^{\min}$ and $\lambda_{\text{peak}}^{\max}$ signify the rigid lower and upper bounds of the peak period tariff, mapping the baseline pricing efficacy and the upper tolerance limit of public expenditure, respectively; similarly, $\lambda_{\text{valley}}^{\min}$ and $\lambda_{\text{valley}}^{\max}$ represent the lower boundary to protect generation-side profit margins and the upper boundary to guarantee sufficient demand stimulation during valley periods, respectively.

Furthermore, to suppress malicious market speculation, mitigate cross-period price arbitrage risks, and eliminate extreme pricing volatility that could trigger systemic market chaos, the peak-to-valley price ratio must be tightly constrained within a verified economic equilibrium corridor. This spatial price-spread constraint is expressed as follows:

$$\gamma = \frac{\bar{\lambda}_{\text{peak}}}{\bar{\lambda}_{\text{valley}}} \quad (16)$$

$$\gamma_{\min} \leq \gamma \leq \gamma_{\max} \quad (17)$$

where γ is the macro peak-to-valley price ratio defined by the average peak tariff $\bar{\lambda}_{\text{peak}}$ and the average valley tariff $\bar{\lambda}_{\text{valley}}$; and $\gamma_{\min}$ and $\gamma_{\max}$ are the lower and upper limits on that ratio. In practice, $\gamma_{\min}$ is usually fixed at 2 so the spread is wide enough to actually shift demand, whereas $\gamma_{\max}$ is capped at 5 so that vulnerable retail customers are shielded from extreme price shocks and the wider market stays stable.

3.2.2. Integrated Network Operational Constraints

Security, reliability, and thermodynamic efficiency in the co-optimized IES all rest on a single constraint framework, which gathers the physical laws, operating envelopes, and technical limits of the electricity distribution grid and the gas transmission network—two systems that are distinct in their physics yet tightly coupled in operation.

The operational safe tracking of the electrical distribution grid is formulated using a generalized branch-flow model. This multi-period framework encompasses distribution network Distflow equations, transmission line thermal boundaries, nodal voltage/current security margins, generator capacity limits, unit commitment and ramping restrictions, and system-wide spinning reserve requirements.

4. Nodal Power Flow and Distflow Constraints

The steady-state multi-period power flow within the distribution network is modeled using the branch-flow model. These equations capture the active and reactive power balance mechanisms coupled with network voltage drop laws:

$$\sum_{j \in \Omega_N} P_{ji,t} - \sum_{k \in \Omega_N} (P_{ik,t} + r_{ik} I_{ik,t}^2) + P_{i,t}^{in} = P_{i,t}^{load} \quad (18)$$

$$\sum_{j \in \Omega_N} Q_{ji,t} - \sum_{k \in \Omega_N} (Q_{ik,t} + x_{ik} I_{ik,t}^2) + Q_{i,t}^{in} = Q_{i,t}^{load} \quad (19)$$

$$V_{j,t}^2 = V_{i,t}^2 - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)I_{ij,t}^2 \quad (20)$$

$$P_{ij,t}^2 + Q_{ij,t}^2 = V_{i,t}^2 I_{ij,t}^2 \quad (21)$$

where $P_{i,t}^{in}$ and $Q_{i,t}^{in}$ denote the active and reactive power inputs injected into node i at time t , respectively; $P_{i,t}^{load}$ and $Q_{i,t}^{load}$ represent the localized active and reactive load demands; $P_{ij,t}$ and $Q_{ij,t}$ indicate the active and reactive power flows traversing from node i to node j ; Ω_N represents the complete set of all electrical nodes in the network; $V_{i,t}^2$ and $V_{j,t}^2$ are the squared voltage magnitudes at nodes i and j ; r_{ij} and x_{ij} denote the inherent resistance and reactance of the transmission line segment connecting nodes i and j ; and $I_{ij,t}^2$ represents the squared current magnitude of line ij at time t .

The power balance conditions represent the strict application of the first law of thermodynamics to maintain system-wide power equilibrium. The voltage-drop relation does the spatial counterpart: it links the voltages of neighboring buses and accounts for the losses and drops that line impedance produces. With the Distflow constraints in force, the grid is held in a secure operating state, resources are placed sensibly across the network, and the chance of a local power mismatch or voltage excursion cascading into wider failure is kept low.

5. Line Transmission Thermal Boundaries

Transmission limits exist to protect the physical line itself. Push the flow past a line's rated capacity and the overcurrent dumps excess Joule heat into the conductor; that heat ages the insulation faster and makes phase-to-ground faults, mechanical damage, or fire more likely. A badly overloaded line also drags down the voltage at downstream nodes, and once that voltage falls below its nominal band, sensitive equipment misbehaves and power quality suffers. Capping the active and reactive flows on each line keeps it clear of thermal overload and preserves a safe transfer margin:

$$-\alpha_{ij}P_{ij}^{\max} \leq P_{ij,t} \leq \alpha_{ij}P_{ij}^{\max} \quad (22)$$

$$-\alpha_{ij}Q_{ij}^{\max} \leq Q_{ij,t} \leq \alpha_{ij}Q_{ij}^{\max} \quad (23)$$

where $P_{ij}^{\max}$ and $Q_{ij}^{\max}$ set the active- and reactive-power thermal ratings of branch ij , respectively, and α_{ij} is a continuous safety-margin coefficient.

6. Nodal Voltage and Current Security Envelopes

Keeping nodal voltages and branch currents inside their local limits is central to grid stability. When voltage sags too low, machines lose electromagnetic torque or are starved

of power, so motors fail to start and control loops behave erratically; when it climbs too high, the insulation in power electronics and transformers is overstressed, which ages it prematurely or breaks it down outright. Currents matter for a different reason. Holding a branch current within its continuous rating limits resistive loss, whereas an unbounded current bleeds energy away as heat, raises operating cost, and shortens the life of the asset:

$$V_{i,\min}^2 \leq V_{i,t}^2 \leq V_{i,\max}^2 \quad (24)$$

$$0 \leq I_{i,t}^2 \leq I_{i,\max}^2 \quad (25)$$

where $V_{i,\min}^2$ and $V_{i,\max}^2$ denote the minimum and maximum allowable operational limits of the squared voltage magnitude at node i , respectively. Similarly, $I_{i,t}^2$ represents the squared current magnitude at node i , restricted by its absolute maximum thermal threshold $I_{i,\max}^2$.

7. Generation Capacity and Ramping Restrictions

The dynamic operational scheduling of generating units is regulated by a combination of capacity boundaries, unit commitment statuses, and ramp-rate limitations. By constraining active power output to a stable operating range and limiting the rate of power adjustment during transitional intervals, these constraints prevent severe mechanical fatigue and thermal stress from aggressive load-following maneuvers. This extends the service life of dispatchable units and ensures continuous, high-fidelity power tracking across diverse grid conditions:

$$u_{i,t}P_i^{\min} \leq P_{i,t} \leq u_{i,t}P_i^{\max} \quad (26)$$

$$(u_{i,t} - u_{i,t-1})R_i^{SU} + (1 - u_{i,t})P_i^{\max} \geq P_{i,t} - P_{i,t-1} \quad (27)$$

$$(u_{i,t-1} - u_{i,t})R_i^{SD} + (1 - u_{i,t-1})P_i^{\max} \geq P_{i,t-1} - P_{i,t} \quad (28)$$

$$-R_i^{down} \leq P_{i,t} - P_{i,t-1} \leq R_i^{up} \quad (29)$$

$$(T_{i,t-1}^{on} - T_{i,\min}^{on})(u_{i,t-1} - u_{i,t}) \geq 0 \quad (30)$$

$$(T_{i,t-1}^{off} - T_{i,\min}^{off})(u_{i,t} - u_{i,t-1}) \geq 0 \quad (31)$$

where $u_{i,t}$ is a binary variable representing the unit commitment status, equaling 1 when the unit is online and 0 when offline. $P_{i,t}$ denotes the active power output of the generator, bounded by its maximum allowable capacity $P_i^{\max}$ and minimum stable generation limit $P_i^{\min}$. The parameters $T_{i,t}^{on}$ and $T_{i,t}^{off}$ are state timers tracking the consecutive operational or non-operational hours of the unit, bounded by the minimum up-time $T_{i,\min}^{on}$ and minimum down-time $T_{i,\min}^{off}$. Lastly, R_i^{SU} and R_i^{SD} denote the start-up and shut-down ramping constraints, while R_i^{up} and R_i^{down} represent the maximum upward and downward ramp-rate limits during normal continuous operation.

8. System Spinning Reserve Capacity Constraints

Reserve capacity constraints dictate that at any given hour within the dispatch horizon, the cumulative residual generating capacity of all online synchronized units must strictly exceed the system's prevailing spinning reserve demand. This headroom is what lets the IES ride out random disturbances. Real grids are routinely hit by the unexpected—a unit trips, or load jumps—and with too little spinning reserve the balance breaks immediately, forcing load shedding or, worse, a cascading blackout that takes economic and social costs with it:

$$\sum_{i \in \Omega_G} (u_{i,t}P_i^{\max} - P_{i,t}) \geq R_t^{req} \quad (32)$$

where Ω_G denotes the complete set of dispatchable generators within the electrical network, and R_t^{req} represents the total spinning reserve capacity required by the system at time t .

The natural gas network utilizes high-pressure pipelines to connect diverse supply, storage, and consumption nodes. To ensure stable and predictable aerodynamic transmission, the network operation must satisfy nodal mass balance, source production limits, nodal pressure envelopes, and nonlinear pipeline fluid flow equations.

9. Nodal Gas Mass Balance Constraints

Based on the principle of mass conservation, the nodal gas balance constraint maintains continuous, stable aerodynamic equilibrium throughout the natural gas network. This constraint requires that, at any given time, the total volume of natural gas injected into a specific node must equal the sum of the outgoing gas flow and the nodal consumption at that node. This balance maintains the operational stability of the gas network:

$$\sum_{s \in \Omega_S} G_{s,t} + G_{P2G,t} + \sum_{j \in \Omega_M} G_{ji,t}^{in} = \sum_{k \in \Omega_M} G_{ik,t}^{out} + G_{i,t}^{load} + G_{i,t}^{gen} \quad (33)$$

where $G_{s,t}$ and $G_{P2G,t}$ represent the natural gas outputs from the traditional gas source s and the P2G facility at time t , respectively; $G_{ji,t}^{in}$ and $G_{ik,t}^{out}$ denote the aggregate gas inflow and outflow volumes at node i coming from or heading to connected nodes; $G_{i,t}^{load}$ is the standard natural gas load demanded at the node; $G_{i,t}^{gen}$ represents the natural gas consumption rate required by gas-fired thermal power units; Ω_S and Ω_M represent the sets of gas sources and gas network nodes, respectively.

10. Gas Source Production and Extraction Limits

Due to geological properties and structural limits of the extraction facilities, the maximum production rate at each gas source is physically capped. At the same time, to prevent reservoir damage, maintain sustainable extraction, and ensure pressure stability at the wellhead, a minimum production threshold is also enforced. These supply limits keep gas production within a sustainable range, preventing over-extraction while securing the system against fuel shortages:

$$G_s^{\min} \leq G_{s,t} \leq G_s^{\max} \quad (34)$$

where $G_s^{\min}$ and $G_s^{\max}$ represent the lower and upper boundary production limits of the gas source s , respectively.

11. Nodal Gas Pressure Security Envelopes

The aerodynamic pressure at all nodes within the network must be maintained within a strictly defined safe range. Structural over-pressure conditions can cause high material stress on the pipelines, increasing the risk of leaks or catastrophic ruptures. Conversely, severe under-pressure conditions can cause flow stagnation or backflow, disrupting gas delivery to end-users and downstream gas-fired generators. Enforcing strict upper and lower pressure thresholds ensures the network operates within a safe range, safeguarding the physical infrastructure:

$$\pi_i^{\min} \leq \pi_{i,t} \leq \pi_i^{\max} \quad (35)$$

where $\pi_i^{\min}$ and $\pi_i^{\max}$ denote the lower and upper operational bounds of the nodal gas pressure, respectively, and $\pi_{i,t}$ represents the actual pressure value at the pipeline node i at time t .

12. Pipeline Flow and Nonlinear Pressure-Drop Constraints

Gas flows along a pipeline because of the pressure difference between its two ends, and that flow follows the Weymouth equation, which is strongly nonlinear:

$$F_{ij,t} = C_{ij} \cdot \text{sgn}(\pi_{i,t} - \pi_{j,t}) \sqrt{|\pi_{i,t}^2 - \pi_{j,t}^2|} \quad (36)$$

where $F_{ij,t}$ denotes the average natural gas flow rate through the pipeline connecting nodes i and j . The pipeline transmission coefficient C_{ij} is a constant value determined by the gas compressibility factor, internal pipeline diameter, gas temperature, friction/efficiency coefficients, and total pipeline length. The mathematical operator sgn represents the sign function, which dictates the physical direction of the gas flow relative to the nodal pressure differential.

Left in this form, the equation is non-convex, so to keep the problem solvable on a commercial MILP solver the squared-pressure terms and the square-root are approximated by piecewise-linear segments, which turn the Weymouth relation into a series of straight pieces.

In this work the squared-pressure difference across each pipeline is taken as the independent variable, and the Weymouth relation is approximated by a piecewise-linear function using twelve segments per pipeline with breakpoints distributed uniformly over the feasible squared-pressure range; a pair of complementary binary variables encodes the sign function so that the direction of flow remains consistent with the nodal pressure differential. With this resolution the maximum absolute deviation between the piecewise-linear approximation and the exact nonlinear Weymouth flow stays below 1.0% across all pipelines, which keeps the gas-flow representation accurate while retaining a mixed-integer linear structure solvable by a commercial solver.

3.2.3. Renewable Generation and Coupled Accommodation Boundaries

Wind is the main source of renewable energy in this IES and a key route to lower carbon emissions, but its output is set by the weather: it is volatile, intermittent, and cannot be dispatched on command. That uncertainty, spread across both time and space, complicates real-time balancing, distribution-grid stability, and the alignment of flows among the coupled carriers.

For wind to sit inside the day-ahead problem without forcing grid violations, the formulation has to fence in the wind farm's operating range and spell out how it works together with demand-side conversion equipment.

13. Wind Power Output Uncertainty and Curtailment Envelopes

Wind cannot be scheduled the way a thermal unit can, so the power a farm actually injects usually differs from its day-ahead forecast. To guard against both shortfalls and local over-generation, the realized output is held at or below the forecast profile by the inequality below:

$$0 \leq P_{w,i,t}^{\text{act}} \leq P_{w,i,t}^{\text{pre}}, \forall i \in \Omega_W, \forall t \in T \quad (37)$$

where $P_{w,i,t}^{\text{act}}$ denotes the actual active power output injected by wind farm i into the grid at time interval t ; $P_{w,i,t}^{\text{pre}}$ represents the ex-ante deterministic day-ahead forecasted baseline of wind power generation for the corresponding node and period; and Ω_W represents the complete set of all operational wind farms across the system topology.

That bound leaves a usable dispatch window: the operator may still curtail wind to keep the network secure, but the synchronized output can never rise above the aerodynamic energy the rotors actually capture.

To make the distinction between available and dispatched wind explicit, and to respect the physical nature of wind as an exogenous realization, an explicit curtailment variable is introduced. The dispatched wind power injected into the grid is written as the forecast availability minus a non-negative curtailment term, with the curtailment bounded above by the available forecast so that it can never exceed what the rotors capture. The forecast value is an exogenous day-ahead input rather than a decision variable; only the non-negative curtailment and the resulting dispatched injection are optimization variables, and only the dispatched amount enters the nodal power-balance equations. The previous inequality is recovered as the special case in which the curtailment variable absorbs the gap between availability and injection, so that the formulation now unambiguously separates the meteorologically available wind from the operator-dispatched wind.

It should be noted that the wind forecast enters the model as a single deterministic day-ahead profile rather than through scenarios, chance constraints, or a robust uncertainty set. This deterministic representation is a deliberate modeling choice: the objective of the study is to quantify how the coordinated flexibility resources—price-based demand response, bidirectional V2G, P2G conversion, and the electricity–gas coupling—reshape the dispatch, and keeping the problem as a single deterministic MILP isolates the marginal contribution of each mechanism without the confounding effect of a particular uncertainty model. The reported costs and curtailment therefore correspond to the expected forecast condition and should be read as an upper bound on achievable benefit; under realized forecast error the actual benefit would be somewhat lower, and the spinning-reserve terms provide only a static buffer against deviations. Embedding the present coordination logic inside a stochastic or distributionally robust framework that explicitly hedges against wind, price, and EV-behavior uncertainty is a natural and important extension, which we identify as future work.

14. Wind-P2G Cross-Carrier Coupling Mechanism

As wind penetration rises, the surplus that appears during off-peak hours has to be dealt with; left alone, it shows up as curtailment and wasted energy. P2G offers a way out. It takes electricity that would otherwise go unused and turns it into synthetic natural gas (SNG), opening a grid-to-gas path that couples the power and gas networks while giving each some room to operate independently of the other.

By tying the wind farms to the P2G units, the model sets how much power each P2G facility may draw. When wind output runs past what the local distribution grid can host, the P2G plant steps in as a fast, flexible load and soaks up the excess, converting fluctuating electricity into gas that can be stored. Curtailment falls, and the IES runs more efficiently in both energy and cost terms.

When wind is scarce, the same coupling pulls the P2G draw back so that conventional dispatchable units can be freed up to keep power and gas flowing reliably to end-users. The coupling logic and the associated equipment limits are written as follows:

$$P_{P2G,j,t}^{elec} \leq \sum_{w \in \Omega_{W,j}} \left(P_{w,w,t}^{pre} - P_{w,w,t}^{act} \right), \forall j \in \Omega_{P2G}, \forall t \in T \quad (38)$$

$$u_{P2G,j,t} P_{P2G,j}^{min} \leq P_{P2G,j,t}^{elec} \leq u_{P2G,j,t} P_{P2G,j}^{max}, \forall j \in \Omega_{P2G}, \forall t \in T \quad (39)$$

where $P_{P2G,j,t}^{elec}$ represents the active electrical power consumed by P2G facility j at time t specifically dedicated to absorbing surplus wind power; $\Omega_{W,j}$ denotes the sub-set of wind farms physically and topologically coupled with the j -th P2G infrastructure; Ω_{P2G} represents the set of all P2G facilities within the system; $P_{P2G,j}^{min}$ and $P_{P2G,j}^{max}$ represent the minimum stable operational consumption threshold and the rated nominal electrical capacity of the

P2G system, respectively; and $u_{P2G,j,t}$ is a binary variable indicating the operational state of P2G unit j at time t , where 1 denotes active status and 0 denotes shut-down mode.

This formulation mathematically ensures that the P2G equipment operates strictly within its structural constraints while accurately mapping the conversion of electricity into natural gas based on the time-varying availability of curtailed wind assets.

The decision to couple the P2G intake primarily to curtailed wind is a modeling choice grounded in the economic and environmental rationale of P2G rather than a hard physical restriction. Because the round-trip efficiency of the power-to-gas-to-power chain is low, supplying P2G from firm or fossil-backed electricity is rarely economic; the value of P2G in this system lies in converting surplus renewable energy that would otherwise be spilled into storable synthetic gas, thereby reducing both curtailment and emissions. Restricting the P2G draw to the curtailed-wind envelope keeps the dispatch aligned with this rationale and prevents the optimizer from exploiting P2G as an artificial load drawing on conventional generation. Relaxing this coupling so that P2G may also absorb low-price grid electricity is straightforward within the same formulation and would mainly affect the relative sizing of curtailment and gas production.

3.2.4. Distributed Asset Operations and Flexibility Limits

To make use of the IES's demand-side elasticity and its room for multi-carrier coordination, this section lays out the operating physics, technical limits, and cost terms of the distributed resources it relies on. Electrochemical storage, electric vehicles, and the multi-energy conversion units between them are what set the local flexibility margins the system draws on to absorb renewable swings and stay in balance.

15. Behind-the-Meter Electrochemical Energy Storage Systems (EESS)

Distributed EEES units installed at various load buses function as critical intertemporal flexibility providers. By shifting energetic blocks across different scheduling horizons, they smooth net-load profiles and enhance grid reliability. To protect battery chemistry and maintain structural integrity, simultaneous charging and discharging within the same scheduling interval is strictly prohibited. The dynamic active power interfaces, complementary operational states, and state of charge of the EEES are regulated by the following mathematical constraints:

$$0 \leq P_{i,t}^{\text{ch}} \leq u_{i,t}^{\text{ch}} P_{i,\max}^{\text{ch}}, \forall i \in \Omega_{\text{EES}}, \forall t \in T \quad (40)$$

$$0 \leq P_{i,t}^{\text{dis}} \leq u_{i,t}^{\text{dis}} P_{i,\max}^{\text{dis}}, \forall i \in \Omega_{\text{EES}}, \forall t \in T \quad (41)$$

$$u_{i,t}^{\text{ch}} + u_{i,t}^{\text{dis}} \leq 1, u_{i,t}^{\text{ch}}, u_{i,t}^{\text{dis}} \in \{0, 1\} \quad (42)$$

$$\text{SOC}_{i,\min} \leq \text{SOC}_{i,t} \leq \text{SOC}_{i,\max}, \forall i \in \Omega_{\text{EES}}, \forall t \in T \quad (43)$$

$$\text{SOC}_{i,t} = \frac{E_{i,t}}{E_i^{\text{rated}}} \quad (44)$$

$$E_{i,t} = (1 - \sigma_i) E_{i,t-1} + \left(P_{i,t}^{\text{ch}} \eta_i^{\text{ch}} - \frac{P_{i,t}^{\text{dis}}}{\eta_i^{\text{dis}}} \right) \Delta t, \forall t \in \{2, \dots, T\} \quad (45)$$

$$E_{i,1} = E_{i,T} \quad (46)$$

$$\sum_{t=1}^T \left(\frac{\eta_i^{\text{ch}} P_{i,t}^{\text{ch}} + \frac{P_{i,t}^{\text{dis}}}{\eta_i^{\text{dis}}}}{2 \cdot E_i^{\text{rated}} \cdot (\text{SOC}_{i,\max} - \text{SOC}_{i,\min})} \right) \leq K_i^{\max} \quad (47)$$

where $P_{i,t}^{\text{ch}}$ and $P_{i,t}^{\text{dis}}$ denote the actual charging and discharging power vectors of EESS unit i at time t , respectively; $P_{i,\max}^{\text{ch}}$ and $P_{i,\max}^{\text{dis}}$ represent the nominal upper power limits; $u_{i,t}^{\text{ch}}$ and $u_{i,t}^{\text{dis}}$ are mutually exclusive binary indicators governing the operating modes; Ω_{ESS} represents the complete set of integrated EESS units; $\text{SOC}_{i,t}$ represents the time-varying state of charge; $\text{SOC}_{i,\min}$ and $\text{SOC}_{i,\max}$ define the lower and upper security boundaries, which standardly calibrated as 5% and 95%; $E_{i,t}$ is the internal energy stored within the cells; E_i^{rated} denotes the nominal rated energy capacity; σ_i is the self-discharge rate; η_i^{ch} and η_i^{dis} represent the round-trip charging and discharging efficiencies; and Δt is the unit dispatch step. The final boundary condition forces cyclical energy conservation to ensure sustainable multi-day scheduling, while $K_i^{\max}$ is a predefined parameter regulating the maximum allowable equivalent full cycles allowed per day to manage capacity fade and structural aging driven by cyclic operation.

16. Vehicle-to-Grid (V2G) Aggregation and Mobility Tracking

Electric vehicles (EVs) represent highly dynamic distributed storage blocks. When aggregated at charging stations, they can engage in flexible V2G energy exchanges, serving as a valuable supplement to traditional demand response programs. The structural power capacity and aggregate energy margins of an EV cluster are non-deterministic and are dictated by the stochastic mobility patterns of vehicle owners. The arrival and departure timelines follow distinct normal probability density functions derived from empirical datasets, which dictate the uncoordinated charging profile and the coordinated multi-period trajectory optimization windows:

$$f(t_{\text{arr}}) = \frac{1}{\sigma_{\text{arr}}\sqrt{2\pi}} \exp\left(-\frac{(t_{\text{arr}} - \mu_{\text{arr}})^2}{2\sigma_{\text{arr}}^2}\right) \quad (48)$$

$$f(t_{\text{dep}}) = \frac{1}{\sigma_{\text{dep}}\sqrt{2\pi}} \exp\left(-\frac{(t_{\text{dep}} - \mu_{\text{dep}})^2}{2\sigma_{\text{dep}}^2}\right) \quad (49)$$

$$\Delta t_n^{\text{ch}} = \frac{(\text{SOC}_n^{\text{exp}} - \text{SOC}_n^{\text{arr}})E_n^{\text{EV}}}{P_n^{\text{ch}}\eta^{\text{EV,ch}}} \quad (50)$$

$$P_t^{\text{EV,tot}} = \sum_{n \in \Omega_{\text{EV},t}} P_n^{\text{ch}} \quad (51)$$

$$\text{SOC}_{n,t} = \text{SOC}_{n,t-1} + \left(\frac{P_{n,t}^{\text{EV,ch}}\eta^{\text{EV,ch}} - \frac{P_{n,t}^{\text{EV,dis}}}{\eta^{\text{EV,dis}}}}{E_n^{\text{EV}}} \right) \Delta t \quad (52)$$

$$\text{SOC}_{n,\min} \leq \text{SOC}_{n,t} \leq \text{SOC}_{n,\max} \quad (53)$$

$$0 \leq P_{n,t}^{\text{EV,ch}} \leq v_{n,t}^{\text{ch}} P_{n,\max}^{\text{EV,ch}}, \quad 0 \leq P_{n,t}^{\text{EV,dis}} \leq v_{n,t}^{\text{dis}} P_{n,\max}^{\text{EV,dis}} \quad (54)$$

$$v_{n,t}^{\text{ch}} + v_{n,t}^{\text{dis}} \leq 1, \quad v_{n,t}^{\text{ch}}, v_{n,t}^{\text{dis}} \in \{0, 1\} \quad (55)$$

$$P_t^{\text{V2G,ch}} = \sum_{n \in \Omega_{\text{EV},t}} P_{n,t}^{\text{EV,ch}}, \quad P_t^{\text{V2G,dis}} = \sum_{n \in \Omega_{\text{EV},t}} P_{n,t}^{\text{EV,dis}} \quad (56)$$

where t_{arr} and t_{dep} denote the vehicle arrival and departure times with statistical parameters configured as $\mu_{\text{arr}} = 17.47$, $\mu_{\text{dep}} = 8.58$, $\sigma_{\text{arr}} = 3.41$, and $\sigma_{\text{dep}} = 3.25$; Δt_n^{ch} represents the continuous uncoordinated charging window for the n -th vehicle; E_n^{EV} is the battery capacity; P_n^{ch} is the constant nominal charging power; and $P_t^{\text{EV,tot}}$ is the total uncoordinated baseline load across the active vehicle set $\Omega_{\text{EV},t}$. The arrival state $\text{SOC}_n^{\text{arr}}$ follows a uniform random distribution between $[0.1, 0.4]$, while the target departure state $\text{SOC}_n^{\text{exp}}$ is bounded within $[0.8, 0.9]$. Under coordinated V2G control, $\text{SOC}_{n,t}$ is the real-time state of charge of vehicle

$n; P_{n,t}^{\text{EV,ch}}$ and $P_{n,t}^{\text{EV,dis}}$ denote its active charging and discharging power; $\eta^{\text{EV,ch}}$ and $\eta^{\text{EV,dis}}$ are the charging and discharging efficiencies; $v_{n,t}^{\text{ch}}$ and $v_{n,t}^{\text{dis}}$ are interlocking binary switches preventing simultaneous charge or discharge; and $P_t^{\text{V2G,ch}}$ and $P_t^{\text{V2G,dis}}$ are the collective grid-side demand and injection vectors. User satisfaction is guaranteed by enforcing that the final state of charge at departure matches the customer's expectation.

17. Socio-Economic Asset Valuation via LCOE

To put the long-run economics of these distributed assets on a common footing within the dispatch objective, their lifecycle is valued through the Levelized Cost of Electricity (LCOE). The method discounts every capital and maintenance outlay back to a single baseline and weighs it against the energy the asset delivers over its life:

$$LCOE = \frac{CAPEX + \sum_{y=1}^Y \frac{OPEX_y}{(1+r)^y}}{\sum_{y=1}^Y \frac{E_y^{\text{out}}}{(1+r)^y}} \quad (57)$$

$$CAPEX = C_{\text{pow}}^{\text{inv}} \cdot P^{\text{rated}} + C_{\text{cap}}^{\text{inv}} \cdot E^{\text{rated}} \quad (58)$$

$$OPEX_y = C^{\text{maint}} \cdot P^{\text{rated}} + C^{\text{labor}} \cdot L_y \quad (59)$$

where $LCOE$ is the unit electricity cost (\$/kWh); $CAPEX$ is the total initial investment, divided into fixed power investment coefficients and capacity volume costs, which are denoted as $C_{\text{pow}}^{\text{inv}}$ and $C_{\text{cap}}^{\text{inv}}$; P^{rated} and E^{rated} define the nominal asset sizes; $OPEX_y$ is the operational and maintenance overhead in year y , which combines power-based maintenance costs and human resource overheads, which are denoted as C^{maint} and C^{labor} ; L_y is the annual labor track; E_y^{out} represents the net cumulative electrical output injected back to the system; and r is the localized interest discount factor, calibrated at 8%.

18. Power-to-Gas (P2G) Multi-Energy Couplers

A P2G plant turns surplus electricity into synthetic natural gas by electrolyzing water and then methanating the hydrogen. Doing so ties the power grid to the gas grid and provides storage at scale; when the electricity is renewable, it also cuts carbon while adding flexibility. The steady-state efficiency and the operating limits of the P2G unit are given by:

$$G_{i,t}^{\text{P2G}} = \eta_{\text{P2G}} \cdot H_{\text{GV}} \cdot P_{i,t}^{\text{P2G,elec}} \quad (60)$$

$$P_{i,\min}^{\text{P2G,elec}} \leq P_{i,t}^{\text{P2G,elec}} \leq P_{i,\max}^{\text{P2G,elec}} \quad (61)$$

where $G_{i,t}^{\text{P2G}}$ represents the volumetric flow rate of synthetic natural gas injected into the gas grid by P2G unit i at time t ; η_{P2G} denotes the electricity-to-gas conversion efficiency; H_{GV} is the standard equivalent heat value conversion factor of natural gas; $P_{i,t}^{\text{P2G,elec}}$ represents the active electrical power intake; $P_{i,\min}^{\text{P2G,elec}}$ and $P_{i,\max}^{\text{P2G,elec}}$ define the minimum stable operating turn-down limit and the rated capacity of the facility, respectively.

19. Gas-Fired Thermal Power Units

Gas-fired thermal plants act as rapid-response balancing units within the IES, adjusting their power output to counter renewable energy fluctuations. Rather than model the combustion and expansion thermodynamics in detail, the unit is reduced to a steady-state linear conversion, which keeps the computation light and leaves the model solvable on a commercial MILP solver:

$$P_{i,t}^{\text{GF}} = \eta_i^{\text{GF}} \cdot H_{\text{GV}} \cdot G_{i,t}^{\text{GF}} \quad (62)$$

where $P_{i,t}^{GF}$ represents the active electrical power delivered by gas-fired unit i at time t ; η_i^{GF} denotes the net electrical efficiency of the turbine; and $G_{i,t}^{GF}$ represents the volumetric natural gas fuel input consumed by the combustion chamber.

20. Conventional Generation Unit Economic Cost Profiles

A conventional unit's fuel cost has two parts: a fixed no-load overhead and a variable fuel term. Because losses and falling turbine efficiency build up as output rises, fuel use climbs faster than linearly near the capacity limit, roughly as the square of output. In economic dispatch formulations, this fuel cost curve is standardly approximated as a convex quadratic function to optimize generation profiles within commercial algorithms:

$$C_i(P_{g,i,t}) = a_i P_{g,i,t}^2 + b_i P_{g,i,t} + c_i \quad (63)$$

where $C_i(P_{g,i,t})$ represents the total operational cost incurred by generator i at time t ; $P_{g,i,t}$ denotes the active power output delivered by the unit; and a_i , b_i , and c_i represent the quadratic cost coefficient, linear cost factor, and fixed no-load cost constant, respectively, which are uniquely calibrated to the unique heat-rate characteristics of the generating unit.

4. Case Study

4.1. Basic Data

To rigorously evaluate the mathematical validity and operational efficacy of the proposed multi-energy coordinated optimal scheduling model, a comprehensive simulation study is conducted on an integrated electricity–gas system. The testbed comprises a modified IEEE 33-node distribution network coupled with a 20-node natural gas grid, as structurally illustrated in Figure 3. The joint optimization framework is mathematically modeled and simulated within the MATLAB 2024a environment, utilizing the high-performance commercial solver CPLEX 12.7.1 to solve the formulated MILP problem. Experimental data are provided in Appendix A.

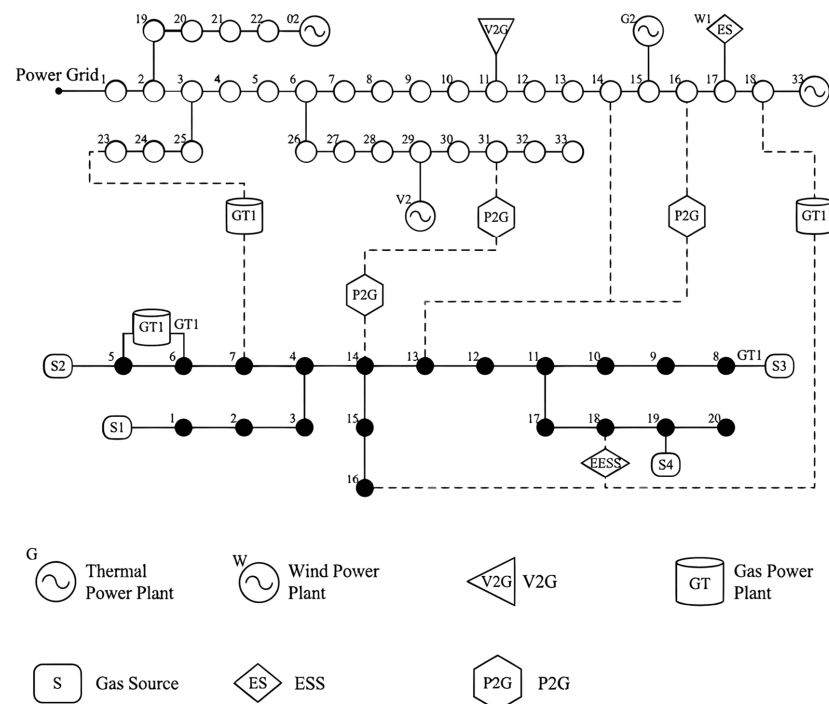


Figure 3. Schematic diagram of IEEE 33-node distribution network and 20-node natural gas grid coupling system.

The localized topological configurations and resource allocations within the coupled network are defined as follows:

Electrical Distribution Subsystem: The baseline voltage of the network is set to 12.66 kV. Node 1 is designated as the slack bus, providing the PCC directly to the upper-level transmission grid. Nodes 18 and 33 are configured as PV nodes, whereas all remaining buses are treated as PQ nodes. Line shunt admittance is neglected, focusing solely on the voltage drops induced by line resistance and reactance. In terms of asset deployment, an EV charging station is integrated at Node 11, and a battery ESS is installed at Node 17. Conventional thermal power plants are situated at Nodes 18 and 22, respectively. Renewable generation resources are represented by two wind farms connected to Nodes 15 and 29. Inter-network energy conversion is facilitated by gas turbines situated at Nodes 25 and 33, which draw fuel from the gas network to inject real power. Conversely, P2G facilities are deployed at Nodes 16 and 31 to transform surplus electricity into SNG for gas network injection. The spatial distribution of baseline active and reactive loads across the electrical nodes is illustrated in Figure 4.

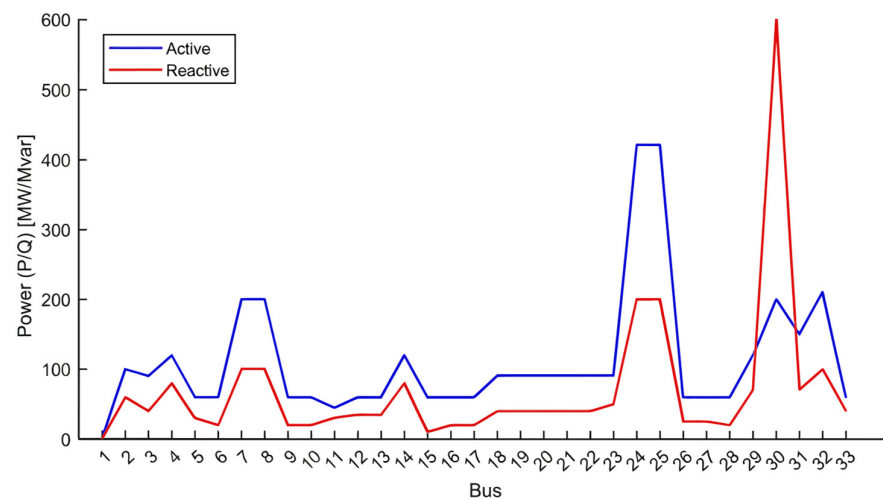


Figure 4. Distribution of active and reactive power of nodes.

Natural Gas Subsystem: Nodes within the gas network are categorically bifurcated into F-nodes and slack nodes. Specifically, Nodes 1, 2, 5, 8, 13, and 14 operate as F-nodes, while the remaining nodes serve as pressure-regulating slack nodes. Natural gas supply sources are positioned at Nodes 1, 5, 8, and 9. The gas consumption nodes for GT generation are located at Nodes 7 and 16, while the SNG streams synthesized by the P2G units are injected into the gas grid at Nodes 13 and 14, thereby establishing a bidirectional energy loop.

Electric Vehicle Aggregator and Scheduling Horizons: Node 11 hosts a charging hub of 1000 EVs, every one of them capable of bidirectional V2G transfer. The schedule spans $T = 24$ h at hourly resolution. To keep the computation manageable while preserving statistical behavior, the fleet is treated as homogeneous in battery chemistry, capacity, and SOC dynamics, with each vehicle's arrival and departure drawn from stochastic distributions.

Treating the fleet of 1000 vehicles as homogeneous in battery capacity, charge/discharge power, efficiency, and SOC dynamics, with universal bidirectional capability, simplifies the analysis but tends to give an optimistic estimate of the dispatchable EV flexibility. Real fleets are heterogeneous in battery size, charger rating, arrival and departure times, target SOC, and willingness to participate, and only a fraction of vehicles are actually V2G-capable. This heterogeneity reduces the usable flexibility through three mechanisms: staggered arrival/departure windows and diverse SOC requirements mean fewer vehicles are simultaneously plugged in with spare capacity to discharge; a partial V2G participation

rate and limited charger availability scale the aggregate discharge capability down roughly in proportion; and once battery-degradation cost is internalized the optimizer discharges more conservatively. The three-case comparison in Section 4.2.4 quantifies the aggregate contribution of the bidirectional V2G fleet, and the dispatchable benefit would diminish in proportion as the genuinely available and V2G-capable share of the fleet decreases.

Case Design: To quantitatively dissect the multi-faceted impacts of DR and V2G spatial–temporal flexibility on IEGS dispatch, three comparative scenarios are executed, as detailed in Table 2:

Table 2. Composition of three cases (✓ = components included, × = components excluded).

Case	Case 1	Case 2	Case 3
PDR	×	✓	✓
V2G	×	×	✓

Case 1 (Baseline): Devoid of PDR mechanisms and lacking V2G bidirectional capabilities, which means EVs function purely as inflexible, unidirectional charging loads.

Case 2 (Demand Response Only): Incorporates PDR but excludes V2G support.

Case 3 (Coordinated Co-Optimization): Runs price-based demand response and bidirectional V2G storage together.

4.2. Results Analysis

4.2.1. Analysis of Electrical Subsystem

Figures 5–7 trace the daily active-power output of the grid interface, the conventional thermal units, and the gas turbines across the three cases.

In the Case 1 baseline of Figure 5, every unit’s output swings through the day in step with the consumer load curve. Through the morning (10:00–13:00) and evening (17:00–20:00) peaks, with no demand-side elasticity to draw on, the system leans hard on the upper-level grid to stay balanced. The local thermal units and gas turbines follow fixed, pre-set schedules with little freedom to move—the classic case of generation chasing load.

Adding PDR in Case 2 reshapes how the units share the load over the day. Responding to TOU price signals, users move shiftable and deferrable load out of the peaks and into the off-peak valleys. Flattening the net-load spikes this way means the import from the upper-level grid ramps up far more gently through the peak than it did in Case 1, and at the same time the local units throttle back during peaks, taking pressure off the system.

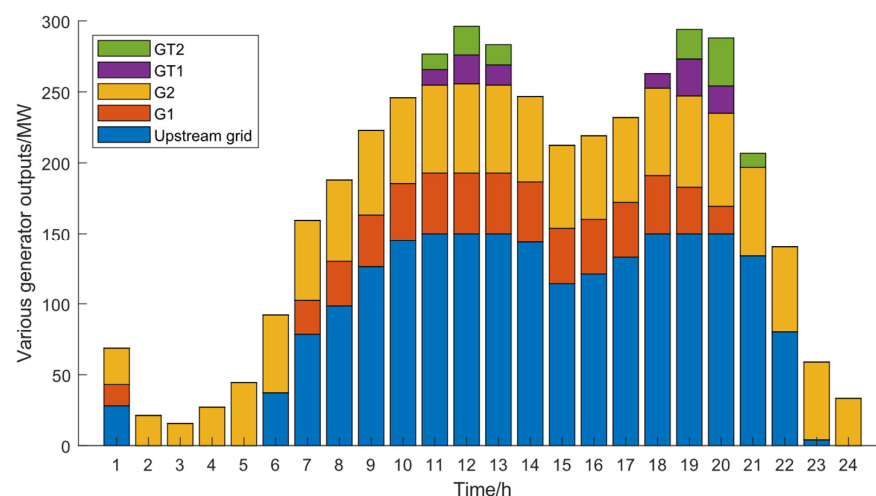


Figure 5. Active power output data of power plant in Case 1.

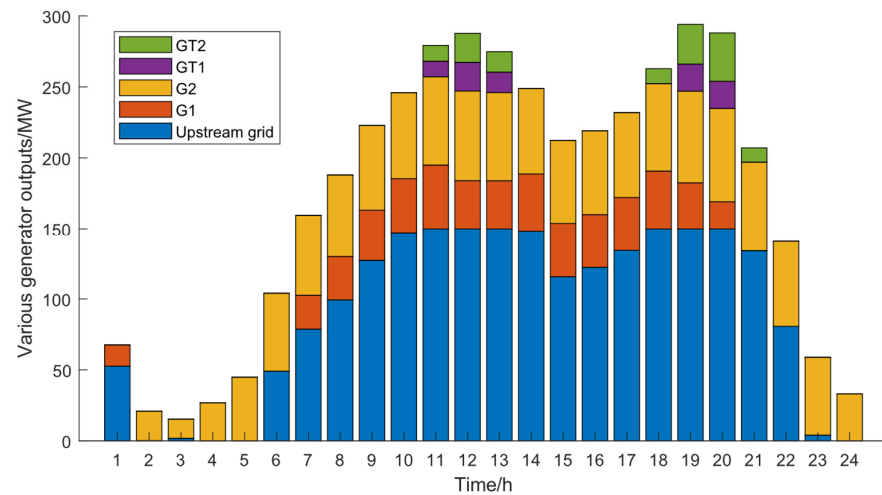


Figure 6. Active power output data of power plant in Case 2.

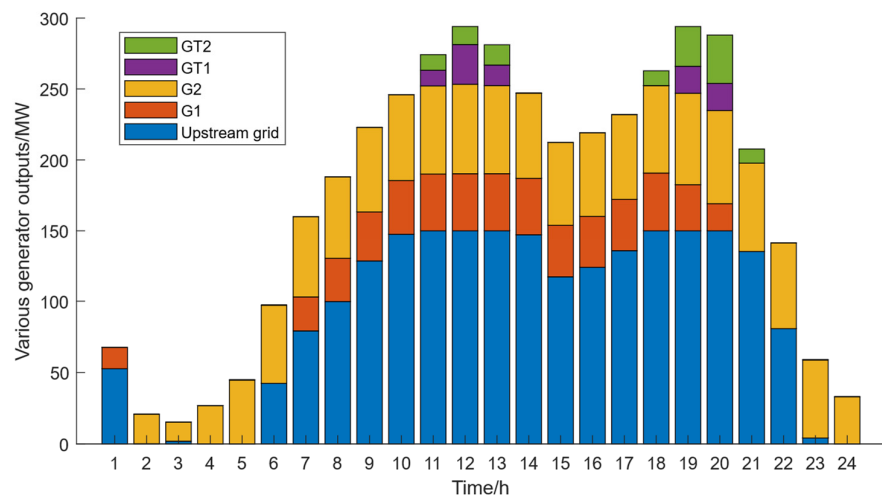


Figure 7. Active power output data of power plant in Case 3.

In Case 3, where PDR is co-optimized with bidirectional V2G car–network interaction, the multi-energy system’s flexibility margins are maximized. As visualized in Figure 7, aggregated EVs act as distributed mobile energy storage resources, feeding power back into the distribution network during peak periods and transitioning into controllable loads during valleys. Consequently, the external power dependency on the upper-level grid during peak hours is further reduced. Meanwhile, local gas turbines synchronize with V2G feedback loops to reshape their generation schedules, scaling back production during peak electricity periods to alleviate gas network stress, and ramping back up to efficient baseload zones during valleys to recharge EV batteries.

4.2.2. Analysis of Gas Subsystem

The multi-energy coupling dynamics between the electrical grid and the natural gas infrastructure dictate the day-ahead supply profiles of the distinct natural gas sources. In Case 1, shown as Figure 8, the natural gas network operates under a rigid, demand-following state where fuel delivery is heavily stressed by the unconstrained electrical peak load. During the peak hours of 10 : 00 – 13 : 00 and 17 : 00 – 20 : 00, gas-fired units demand a massive, instantaneous fuel volume, forcing the gas sources to ramp up sharply. Within this network topology, source S_4 functions as the marginal peaking supplier, exhibiting an aggressive delivery surge compared to its overnight baseline, while suppliers S_1 , S_2 , and S_3 operate near their maximum contractual nominations and physical pipeline pressure

boundaries. This rigid coupling demonstrates how unmitigated electrical peaks propagate across network boundaries, threatening the nodal pressure stability and line-pack buffers of the gas infrastructure.

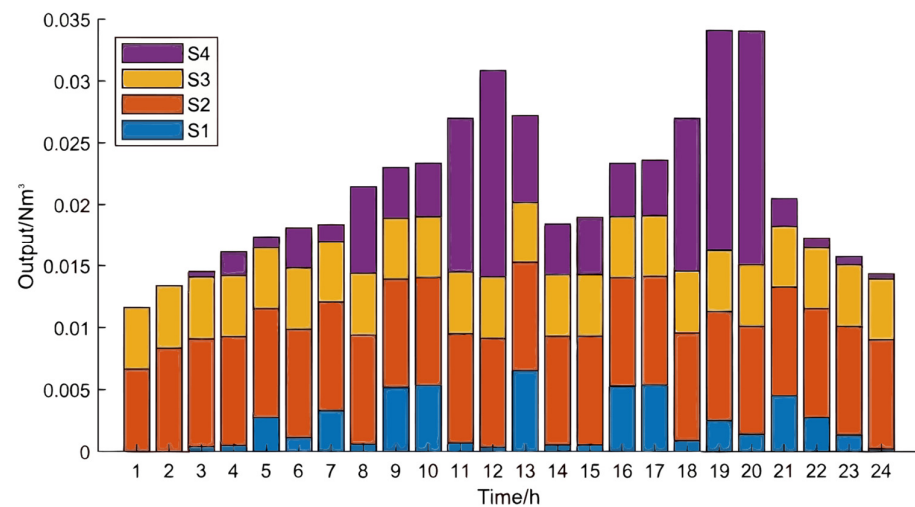


Figure 8. Gas source output data of Case 1.

In Case 2, as detailed in Figure 9, the implementation of PDR relaxes the inter-network operational coupling by flattening the gas infrastructure's supply ramps. Because the electrical TOU price signals smoothly redistribute the load curve, the dispatch of P2G units during grid peaks is minimized, leading to a noticeable decline in delivery rates for major gas sources, particularly S_3 , during high-tariff hours. This coupling relaxation successfully prevents localized pipeline congestion and dynamic pressure drops, yielding a more uniform gas production profile. The minor off-peak supply increments observed further confirm that demand-side flexibility in the power sector can be strategically leveraged to optimize the logistical efficiency and pressure regulation of the coupled gas network.

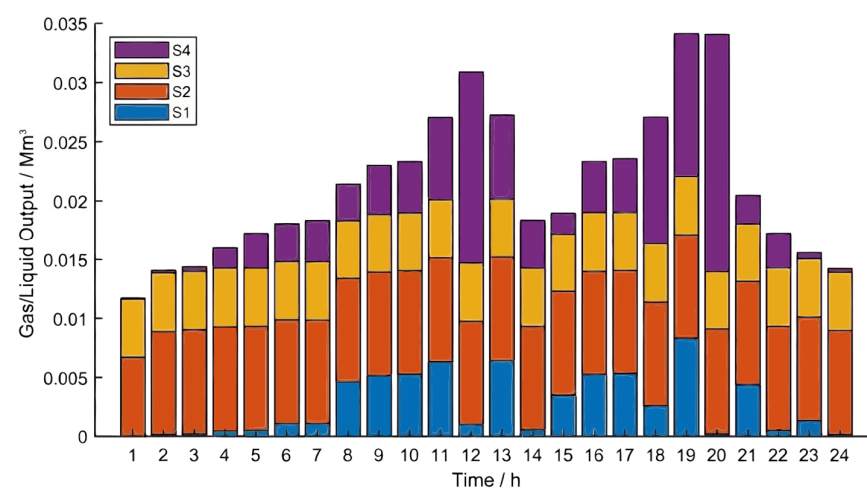


Figure 9. Gas source output data of Case 2.

This multi-energy decoupling mechanism is maximized in Case 3 through the synergistic incorporation of V2G assets and P2G coupling shown as Figure 10. During peak intervals, the localized discharge from vehicle batteries substitutes for gas-turbine peaking capacity, enabling the marginal source S_4 to operate at a significantly reduced, stress-free level. The valley hours then pick up extra load from shifted demand and EV charging, which raises gas use; this happens, though, while line-pack is ample and tariffs are low, so

the gas sources can follow the system's actual state instead of chasing electrical spikes, and the coupled infrastructure stays balanced.

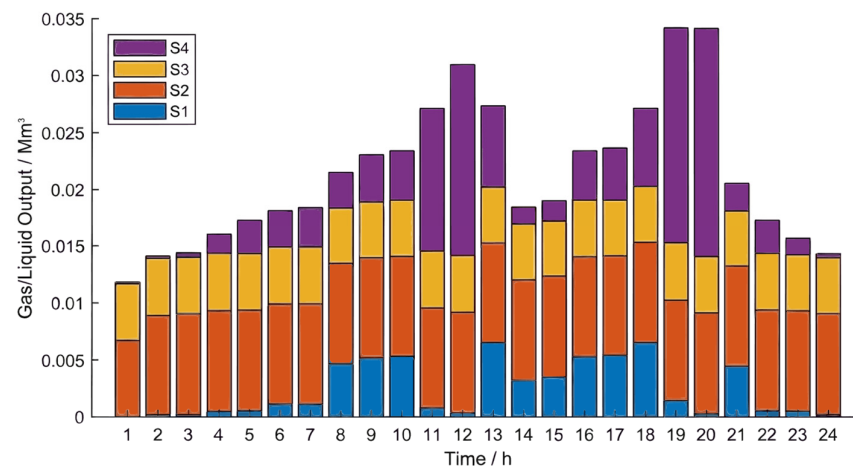


Figure 10. Gas source output data of Case 3.

4.2.3. Analysis of DR Results

Comparing the three cases on operating cost, peak-shaving, and locational marginal price brings the trade-offs into focus in Figure 11. Case 3 sits at the economic and operational frontier, with the lowest total operating cost and the flattest peak-to-valley load curve. Case 2 lands in the middle, and Case 1, with no flexibility assets at all, comes out worst.

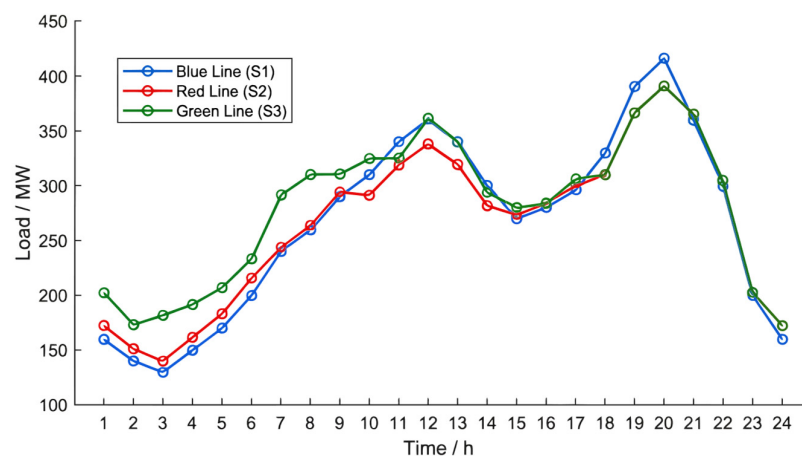


Figure 11. Daily load distribution in different scenarios.

Figure 12 shows how prices are set differs sharply from case to case. In Case 1 they sit on a flat baseline that never signals scarcity, so the load profile comes out raw and unoptimized. In Case 2, PDR narrows the peak-to-valley gap, but with no storage to change the total energy drawn, the local leveled cost stays high and the peak price still spikes. In Case 3, pairing DR with V2G smooths these anomalies: V2G discharge adds local capacity through the peaks to hold the spikes down, while coordinated overnight charging adds load that lifts the valley price above its level in Case 2. That higher off-peak price is useful rather than wasteful—it flattens the curve and spares the generators some ramping wear.

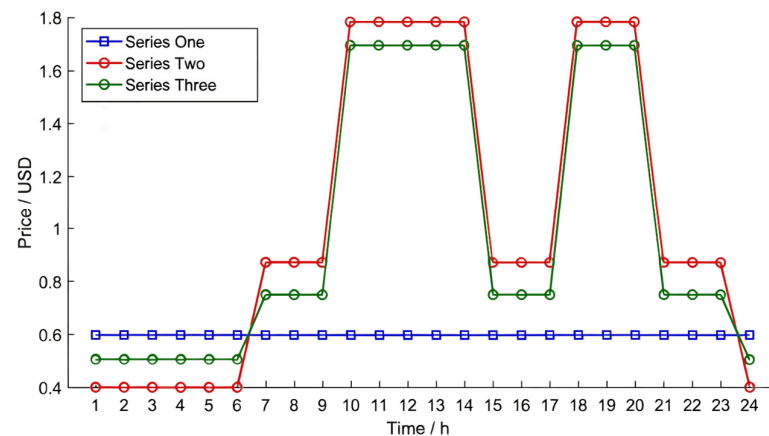


Figure 12. Distribution of daily demand electricity prices under different scenarios.

Taken together, the simulations show that letting demand respond inside the IES schedule brings total operating cost down markedly, and that a well-set TOU tariff flattens the peak-to-valley gap. Adding V2G strengthens the power–gas coupling on both sides, drawing on multi-energy complementarity to steady fuel sourcing and take up intermittent renewable output. The caveat is scale: as large numbers of heterogeneous, controllable loads come online in smart grids, holding a coupled IES stable and reliable will get harder, which points to robust, decentralized, and stochastic dispatch as the work worth pursuing next.

4.2.4. Computational Performance, Sensitivity, and Ablation Analysis

The complete model is a mixed-integer second-order cone program assembled over the 24 h horizon. In the full Case 3 configuration it comprises on the order of fifteen thousand decision variables, roughly four thousand of which are binary, dominated by the piecewise-linear Weymouth segment selectors together with the unit-commitment, V2G, and storage charge/discharge indicators, alongside a comparable number of network and device constraints. Solved with CPLEX 12.7.1 on a standard desktop workstation, each configuration converges to a relative optimality gap below 0.01% within a few seconds of wall-clock time, confirming that the formulation is computationally lightweight at this scale. Solution time grows mainly with the number of Weymouth segments and the granularity of the EV aggregation; for substantially larger multi-area systems a decomposition or clustering scheme would be advisable, which we note as a direction for future scalability studies.

Table 3 summarizes the quantitative comparison of the three operating cases. Relative to the uncoordinated baseline of Case 1, the price-based demand response of Case 2 lowers the total daily operating cost by 3.2% and compresses the peak-to-valley load difference by 15.0%, and the addition of bidirectional V2G in Case 3 deepens these to a 4.7% cost reduction and a 31.9% reduction in the peak-to-valley difference. Wind curtailment stays at zero in all three cases, since the power-to-gas unit converts any surplus wind into synthetic gas rather than spilling it. The dispatch nonetheless remains far more sensitive to wind availability than to the other resources, as the ablation analysis below quantifies.

Table 3. Quantitative comparison of the three operating cases; cost and peak-to-valley reductions are taken relative to the Case 1 baseline.

Case	Total Cost (USD)	Cost Reduction	Peak-to-Valley (MW)	Peak–Valley Reduction	Wind Curtailment
Case 1 (Baseline)	4,873,632	–	239.4	–	0%
Case 2 (PDR)	4,718,301	3.2%	203.5	15.0%	0%
Case 3 (PDR + V2G)	4,644,496	4.7%	163.1	31.9%	0%

To isolate the contribution of each model component, an ablation study was performed in which individual mechanisms are removed from the full Case 3 configuration; the resulting changes in total operating cost and wind curtailment are reported in Table 4. Three observations stand out. First, the P2G unit is the decisive enabler of renewable accommodation: removing it raises the share of available wind that is curtailed from essentially zero to about 14.5%, because the surplus wind that the network cannot absorb can no longer be diverted to synthetic-gas production, while the total cost rises by only about 0.6% since the additional gas must instead be procured from upstream supply. Second, reducing the wind resource by 30% is the most cost-significant change, increasing the total operating cost by about 9.2% and confirming that low-cost wind energy is the primary economic driver of the dispatch. Third, removing the storage-degradation cost term lowers the reported cost by only 0.42% and leaves the remaining cost components essentially unchanged, because the storage already operates at its cycle-count limit in every configuration, so the penalty affects the cost accounting rather than the dispatch; relaxing the gas-network constraints changes the cost by less than 0.01%, indicating that the pipeline-pressure and Weymouth limits are not binding for this test system.

Table 4. Ablation study of model components, relative to the full Case 3 model.

Configuration	Δ Total Cost	Δ Wind Curtailment	Note
Full model (Case 3)	reference	reference	all mechanisms active
Without LCOE degradation	−0.42%	≈0 pts	storage at cycle limit; affects cost accounting only
Without P2G	+0.61%	+14.5 pts	surplus wind no longer absorbed
Without gas-network constraints	≈0%	≈0 pts	gas network not binding in this case
Wind penetration −30%	+9.17%	≈0 pts	more thermal and gas needed; no surplus to curtail

Taken together, these results characterize the marginal role of each flexibility resource in the present system. On the demand side, the price-based demand response alone delivers about a 3.2% reduction in total operating cost and a 15.0% compression of the peak-to-valley load difference relative to the uncoordinated baseline, and adding bidirectional V2G raises these figures to 4.7% and 31.9% respectively, so the V2G layer contributes most of the additional peak-shaving while adding a smaller further cost saving. Power-to-gas is the principal contributor to renewable accommodation: with it active essentially all of the available wind energy that is not injected at the busbar is converted to synthetic gas rather than spilled, whereas disabling it exposes about 14.5% of the available wind as curtailment. The electrochemical storage contributes by shifting energy across the day, but because it operates at its cycle-count limit its marginal cost effect is small. The gas network, while essential for delivering gas-fired generation and the P2G output, does not constrain the optimum at the present loading, so relaxing it leaves the dispatch essentially unchanged. Wind availability is the dominant economic factor, so the cost benefit of the coordinated framework is largest when renewable penetration is high. These proportions are specific to the present test system and would shift with network topology, resource sizing, and price levels.

5. Conclusions

This paper has presented a comprehensive operational optimization and day-ahead scheduling framework for a coupled power–gas IES, systematically examining the synergistic impacts of PDR and V2G integration across three distinct operational scenarios. The empirical simulation results yield several critical techno-economic and structural insights.

- (1) Demand-side flexibility through PDR reshapes the load to a real degree, shaving the sharp peaks and moving elastic load into the off-peak valleys. That eases stress on local infrastructure and cuts how much the system has to lean on costly marginal peakers and on upstream grid imports during high-tariff hours.
- (2) Running V2G together with demand response builds a flexible, decentralized storage layer that firms up the power–gas coupling. Discharging in the peaks and charging in the valleys, the EV fleet acts as a buffer that steadies fuel draw at the gas nodes, damps line-pack swings, and makes room for more intermittent renewables. In cost terms, coordinating DR and V2G reaches the Pareto frontier: the lowest total multi-energy operating cost and, at the same time, the narrowest peak-to-valley load gap.
- (3) The locational marginal prices tell the same story from the market side: demand response on its own runs into a flexibility ceiling, but once V2G is added the peak-price anomalies are damped and the valley tariff is lifted in a useful way, which confirms that the closed-loop multi-energy clearing works as intended.

Quantitatively, on the studied integrated electricity–gas test system the price-based demand response lowers the total daily operating cost by about 3.2% and compresses the peak-to-valley load difference by about 15.0% relative to the uncoordinated baseline, and the further addition of bidirectional V2G deepens these gains to a 4.7% reduction in total operating cost and a 31.9% reduction in the peak-to-valley difference. Wind curtailment remains essentially zero across the three coordinated cases, because the power-to-gas unit converts the surplus wind that the network cannot accommodate into synthetic gas rather than spilling it; without the power-to-gas unit, about 14.5% of the available wind energy would be curtailed.

It should be emphasized that these quantitative findings are obtained on a single modified IEEE 33-node electrical and 20-node gas test system under specific parameter assumptions; the numerical magnitudes should therefore be read as indicative trends rather than universal values, and confirmation on larger and more diverse systems remains necessary.

However, as the diversity and nonlinearity of controllable, heterogeneous loads accelerate in modern smart grids, the operational stability, cyber-physical security, and reliability of coupled IES will face unprecedented stochastic challenges. Consequently, developing robust, decentralized, and distributionally robust optimization strategies capable of accounting for the multi-period uncertainties of both EV driving behaviors and renewable energy availability remains a compelling and vital direction for future scholarly investigation.

Building on these directions, future work will proceed along three concrete lines: incorporating explicit uncertainty modeling for wind output, energy and gas prices, and EV behavior through stochastic or distributionally robust optimization; adding a rolling real-time correction layer on top of the day-ahead schedule to absorb intra-day deviations; and addressing practical implementation aspects such as communication latency, metering accuracy, and the diversity of real EV fleets.

Author Contributions: X.L.: Methodology, Writing—Original draft. P.R.: Methodology, Writing—Original draft. J.C.: Conceptualization, Writing—Review and Editing. R.D.: Writing—Review and Editing. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: Author Xun Lu was employed by the Guangdong Power Grid Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The funder had the following involvement with the study: design, analysis, interpretation of data, and the writing of this article.

Nomenclature

The principal sets, parameters and decision variables used throughout the paper are summarized below; subscripts identifying individual nodes, units or time slots follow the conventions defined where each equation is introduced.

Symbol	Description
Sets and indices	
t, T	time slot and the 24 h scheduling horizon
i, j	electrical/gas network node indices
$\Phi_{\text{CHP}}, \Phi_{\text{GT}}, \Phi_{\text{wind}}$	sets of dispatchable generators and wind farms
$\Phi_{\text{P2G}}, \Phi_{\text{EV}}$	sets of power-to-gas units and electric vehicles
Parameters	
$\Phi_{\text{peak}}, \Phi_{\text{flat}}, \Phi_{\text{valley}}$	time-of-use retail prices for peak, flat and valley blocks
$\varepsilon, \varepsilon_{t,t'}$	self- and cross-price demand elasticities
$p_{k,t}^{\text{wind,fore}}$	day-ahead forecast (available) wind power
$\eta^{\text{P2G}}, \eta^{\text{ev-ch}}, \eta^{\text{ev-dis}}$	P2G conversion, EV charging and discharging efficiencies
$\text{SOC}^{\text{arr}}, \text{SOC}^{\text{dep,tar}}$	EV arrival and required departure states of charge
a_k, b_k, c_k	quadratic, linear and constant generator cost coefficients
Decision variables	
p_t^{grid}	active power exchanged with the upper-level grid
$p_{k,t}^{\text{wind}}$	dispatched and curtailed wind power
$p_{m,t}^{\text{ev-ch}}, p_{m,t}^{\text{ev-dis}}$	EV charging and discharging power (V2G)
$\text{SOC}_{k,t}$	EV state of charge
$p_{k,t}^{\text{P2G}}$	electrical power drawn by P2G units
$Q_{k,t}^{\text{src}}, Q_{k,t}^{\text{P2G-gas}}$	natural gas volumetric flow/production
$L_t^{\text{shift}}, L_t^{\text{cut}}$	shiftable and interruptible load after PDR
$u_{k,t}^{\text{P2G}}, x_{k,t}^{\text{ch}}, x_{k,t}^{\text{disch}}$	binary on/off and charge/discharge state variables

Appendix A

Table A1. Generator equipment data.

Bus Number	Active Power/MW	Reactive Power/MVar	Upper Active Limit/MW	Lower Active Limit/MW	Upper Reactive Limit/MVar	Lower Reactive Limit/MVar	Base Capacity/MVA
1	0	0	1.5	0	2.5	−2.5	100
18	0.2	0	1.5	0.15	1	0	100
22	0.2	0	1.3	0.13	1	0	100
25	0.2	0	1.2	0.1	1	0	100
33	0.2	0	1.3	0.1	1	0	100

Table A2. Generator constraint data.

Bus Number	Min Up Time/h	Min Down Time/h	Initial Up Time/h	Initial Down Time/h	Ramp-Up Rate/(MW/h)	Ramp-Down Rate/(MW/h)
1	1	1	1	0	3	3
18	3	2	2	0	0.3	0.3
22	3	2	0	2	0.26	0.26
25	2	1	0	1	0.24	0.24
33	2	1	0	1	0.26	0.26

Table A3. Generator cost data.

Bus Number	Start-Up Cost/USD	Shut-Down Cost/USD	Quadratic Coefficient	Linear Coefficient	Constant Term
1	0	0	0	70,000	0
18	5000	2000	3500	67,000	300
22	3000	1500	3000	61,807	300
25	2000	1000	2300	65,000	150
33	2000	1000	2300	65,000	150

Table A4. Natural gas network node data.

Bus Number	Load/(Mm ³ /Day)	Pressure/Bar	Maximum Pressure/Bar	Minimum Pressure/Bar
1	0	64.12	77	0
2	0	64.05	77	0
3	5.88	63.76	80	30
4	0	60.41	80	0
5	0	63.05	77	0
6	6.05	58.80	80	30
7	7.88	58.73	80	30
8	0	64.60	66.2	50
9	0	64.03	66.2	0
10	9.55	61.71	66.2	30
11	0	60.52	66.2	0
12	1.71	58.40	66.2	0
13	0	56.76	66.2	0
14	0	56.49	66.2	0
15	10.27	53.65	66.2	0
16	23.42	50.00	66.2	50
17	0	60.16	66.2	0
18	0	57.18	63	0
19	0.33	44.19	66.2	0
20	1.71	43.63	66.2	25

Table A5. Data on air supply.

Gas Source	Supply/(Mm ³ /h)	Maximum Supply/(Mm ³ /h)	Minimum Supply/(Mm ³ /h)	Pressure/Bar	Cost/(USD/m ³)
1	35.83	36.23	0	64.12	0.44
2	24.06	26.25	0	64.05	0.43
3	15	15	0	63.05	0.42
4	54.17	68.79	0	64.60	0.44

Table A6. Vehicle parameters.

Vehicle Parameter	Value
Battery Capacity/kWh	60
Rated Charging Power/kW	7
Rated Discharging Power/kW	7
Desired User SOC Value	0.9
Charging Efficiency	90%
Discharging Efficiency	90%

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